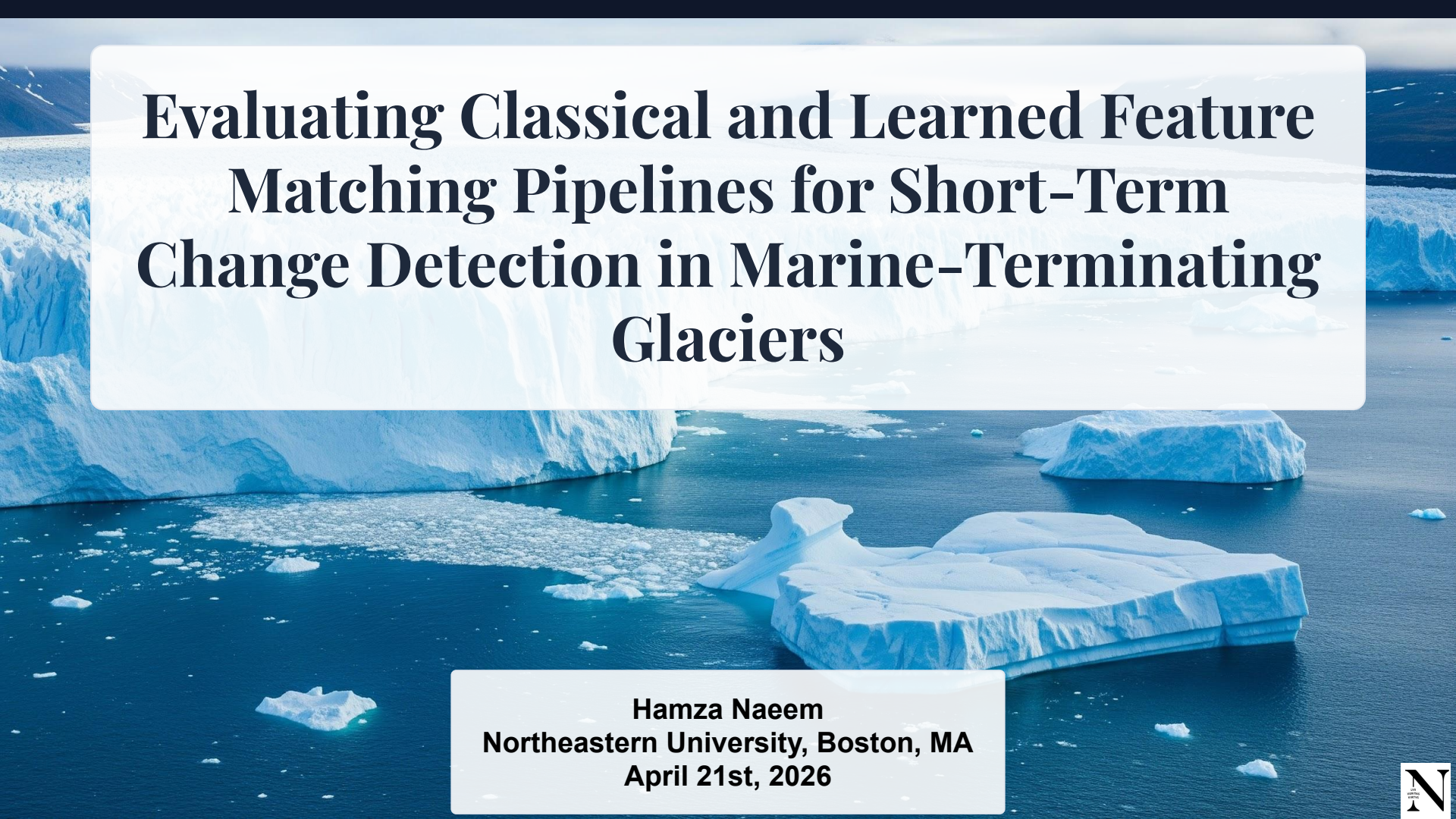
Evaluating Classical and Learned Feature Matching Pipelines for Short-Term Change Detection in Marine-Terminating Glaciers



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April 21st, 2026

Presentation Chapters

Chapter 1: SFM Pipelines

Evaluation of Classical (Geometric) and Learned (ML-Based) Structure-from-Motion (SfM)Pipelines

- The Core Problem & Objectives
- Datasets Analysis: Why this data?
- Pipeline Architecture & Design
- Integrating Dense Matchers
- Key Results
- Comparison

Chapter 2: Change Detection

Application in Marine-Terminating Glaciers

- Alignment Strategies (GPS vs. Feature-Based)
- Teaser++ Integration
- Change Detection Methodologies
- Accuracy Assessment
- Optimal Detector & Model Selection
- Key Results

The Problem & Usecases

Autonomous robots — ROVs, AUVs, UAVs — rely on **feature matching** as the first stage of almost every visual perception task:

SLAM — Localization & Mapping

SfM — 3D Reconstruction

VPR — Visual Place Recognition

1. **Life Evolution in Deep Seas**
2. **Change Detection in Glaciers**
   a. **Sea Level Rise & Global Warming**
   b. **Track Climate Change**
3. **Advancing Perception in Harsh Areas like Sea, Space, Low Texture**

1. **Which descriptor** do you actually trust when the scene breaks the assumptions your paper benchmarks were designed for?
2. And **what does it cost** to run a dense matcher on 100+ images from a ship-side workstation, vs running SIFT on the same data?
3. Should you use dense matchers for change detection?

These questions are not answered by HPatches, ScanNet, or MegaDepth datasets.

Real deployments hit **unseen** scenes. Turbid water, ice, biological clutter, poor lighting — none of these are in the standard benchmarks.

Datasets



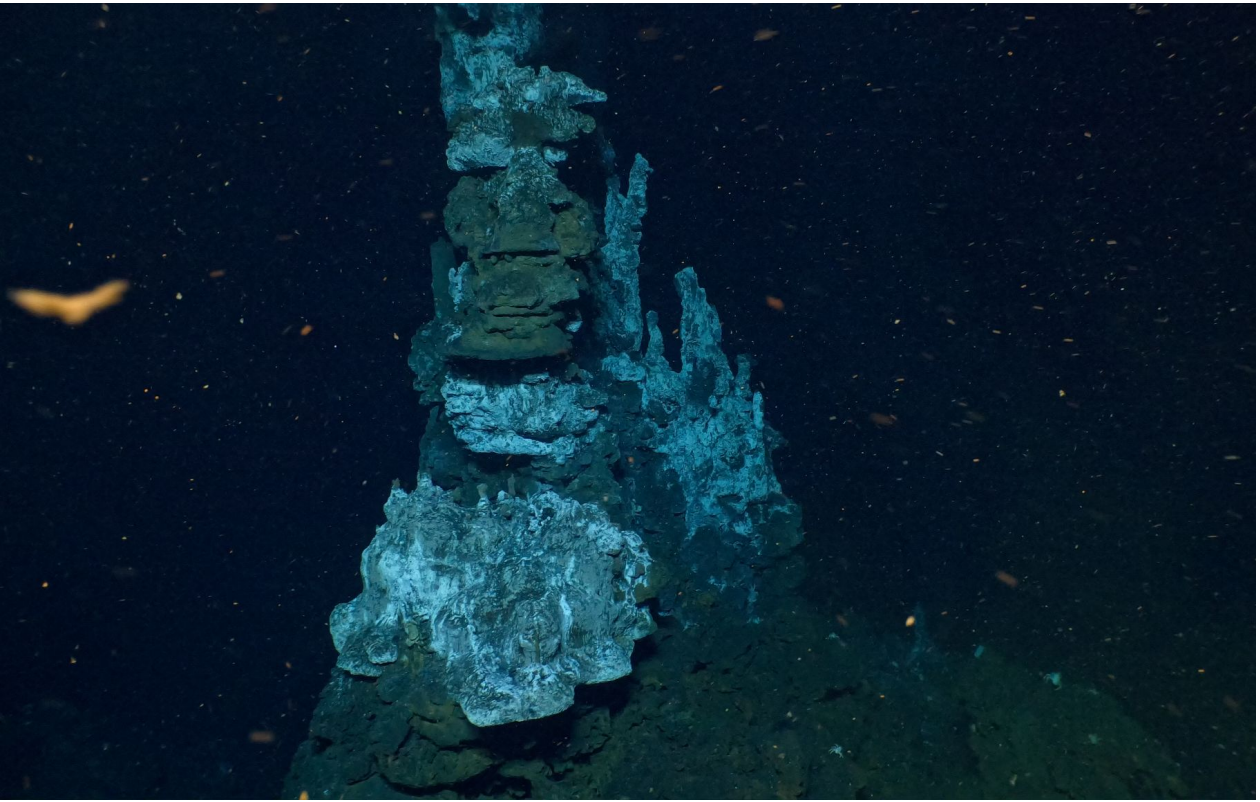
Glacier

66 images, Aerial

- Svalbard (Arctic Ocean)
- Low-texture ice
- Self-similar crevasses
- Non-Lambertian snow
- Two Flights, 1 day apart

Canonical failure case for classical descriptors

Datasets



Hydrothermal Vent

108 images, Deep-Sea

- Turbidity & backscatter
- Biological clutter
- Artificial lighting shadows
- **Bio9 site**
- 2,500 meters deep in the Pacific Ocean

Datasets



UAV City Hall

65 images, Aerial Orbit

- Strong texture
- Consistent geometry
- Heritage3DMT dataset

Controlled baseline (easy scene)

Pipeline

Approach: A Unified SfM Pipeline

Plug every descriptor into the **same COLMAP pipeline**. Compare reconstruction quality on identical data.



What changes

Only **stages 1 & 2** — the descriptor/matcher pair.

What stays constant

- Image resolution (1600 px)
- Max 8000 matches/pair
- RANSAC threshold (4 px)
- Exhaustive Matching
- COLMAP mapper defaults

Integrating Dense Matchers

Dense matchers (LoFTR, DKM, RoMa) produce **per-pair** pixel correspondences — no consistent keypoint IDs. To plug into COLMAP, we cluster them into canonical per-image keypoints.

Dense matches per pair – **Stream to HDF5** – **Radius-cluster per image (3 px)** – **Re-index matches → COLMAP DB**

HDF5 Streaming Strategy

HDF5 streaming is critical — with 108 images exhaustive = ~5,778 pairs × 8k matches, holding all correspondences in RAM isn't feasible.

We stream, compress (gzip), and read back per-image to ensure scalability and memory efficiency.

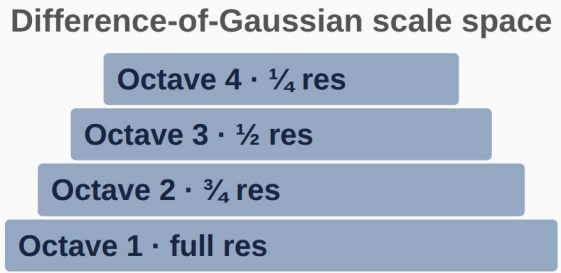
Evaluated Pipelines

Pipeline	Type	Output	Matcher
SIFT	Classical	Sparse	Guided matching
ORB	Classical	Sparse	BF + cross-check
AKAZE	Classical	Sparse	BF + ratio test
SuperPoint + SuperGlue	Learned	Sparse	GNN + Sinkhorn
ALIKED	Learned	Sparse	LightGlue
DISK + LightGlue	Learned	Sparse	LightGlue
LoFTR	Learned	Semi-dense	Coarse-to-fine
DKM	Learned	Dense	Kernel regression
RoMa	Learned	Dense	Warp estimation

SIFT Classical Sparse

Lowe, 2004 · "Distinctive Image Features from Scale-Invariant Keypoints"

Difference-of-Gaussian scale space



Keypoints = extrema across scale ⊕ space
Descriptor = 128-d gradient histogram

Keypoints = extrema across scale & space
Descriptor = 128-d gradient histogram

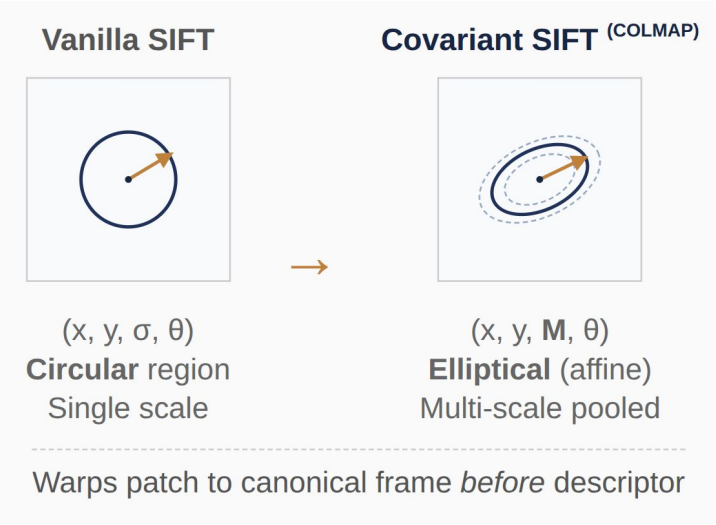
Key idea

- Detect at DoG extrema → scale-invariant keypoints
- 128-dim float descriptor from oriented gradients
- COLMAP adds guided matching (F-matrix aware)

"Still the gold standard when texture is usable"

SIFT vs Covariant SIFT Classical

What COLMAP actually runs: SIFT + affine-shape adaptation + domain-size pooling



Two additions on top of SIFT

1. Affine-shape adaptation (Mikolajczyk & Schmid 2004)

- Estimates a 2×2 affine shape matrix M per keypoint
- Warps elliptical region → canonical frame
- Invariant to viewpoint slant / perspective distortion

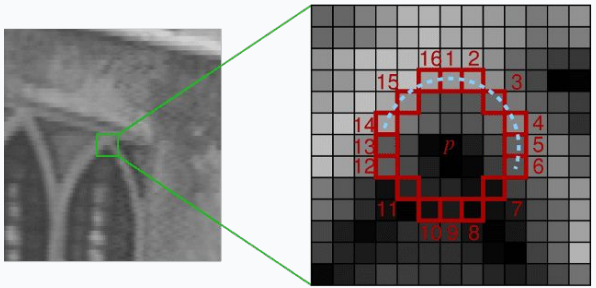
2. Domain-Size Pooling (DSP-SIFT) (Dong & Soatto 2015)

- Pools gradient contributions across multiple scales
- Robust to scale-estimation errors
- Empirically stronger than vanilla SIFT on wide-baseline pairs

*This is why SIFT is **still competitive** on our glacier and UAV datasets — it's not vanilla 2004 SIFT.*

ORB / AKAZE Classical Sparse

ORB: Oriented FAST and Rotated BRIEF



Key idea

- Oriented FAST detector (very fast)
- Rotated BRIEF binary descriptor (32 bytes)
- Config: WTA_K=3, 12 pyramid levels, cross-check

"Binary descriptors can't handle turbid / low-texture scenes"

AKAZE: Fast Explicit Diffusion

Nonlinear scale space preserves edges



SIFT (Gaussian)
edges blur

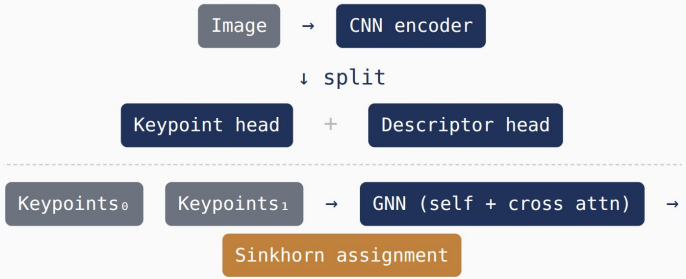
AKAZE (nonlinear)
edges preserved

Key idea

- Perona-Malik diffusion (non-Gaussian)
- Fast Explicit Diffusion (FED) → near-ORB speed
- MLDB binary descriptor

SuperPoint + SuperGlue Learned Sparse

DeTone 2018 + Sarlin 2020: Learned detector + graph-neural matcher

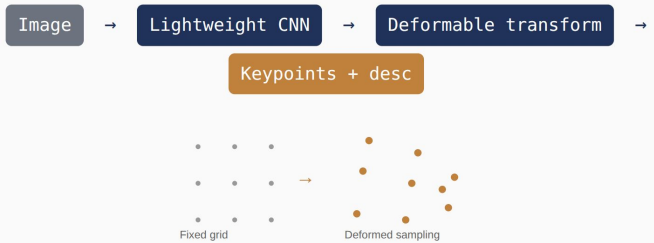


SuperPoint trains via homographic adaptation
SuperGlue solves matching as an optimal transport problem

Key idea

- Joint detection + description from a single CNN
- GNN matcher uses **global context** (all-to-all attention)
- Sinkhorn iterations enforce partial-assignment constraint

ALIKED Learned Sparse



Samples adapt to local image content → spatially-aware descriptors

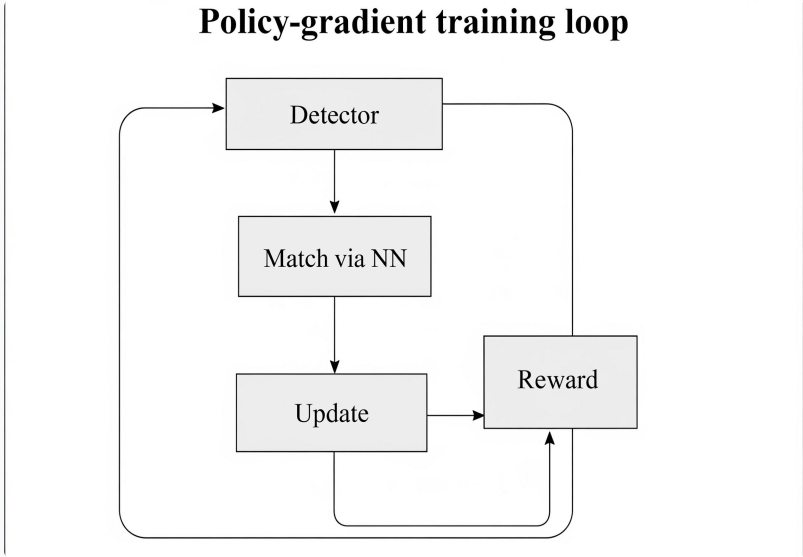
Key idea

- Deformable sampling layer (learned offsets)
- Neural reprojection error supervision
- Matched with LightGlue in our pipeline

Best sparse quality/compute trade-off on harsh scenes

DISK + LightGlue Learned Sparse

Tyszkiewicz 2020 + Lindenberger 2023 · RL-trained detector + Adaptive attention



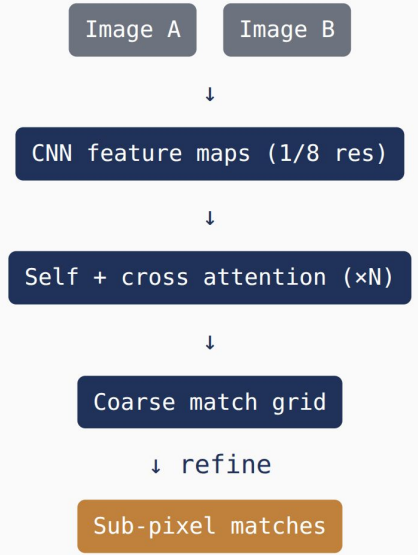
Key idea

- DISK: reward = number of correct matches → trains for matchability
- LightGlue: adaptive depth, skips unnecessary layers
- Faster than SuperGlue at similar quality

"Fastest learned method — great for GPU-constrained payloads"

LoFTR Learned Semi-dense

Sun et al., 2021 · "Detector-Free Local Feature Matching with Transformers"



No keypoint detection — matches **feature maps** directly

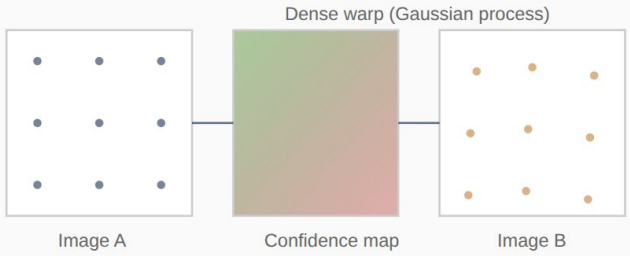
Key idea

- Skip the detector — match feature-map cells directly
- Transformer layers aggregate long-range context
- Coarse-to-fine: 1/8 res → sub-pixel refine

Strong on repetitive / low-texture — but slow

DKM Learned Dense

Edstedt et al., 2023 · "Dense Kernelized Feature Matching for Geometry Estimation"



Every pixel → correspondence + reliability score

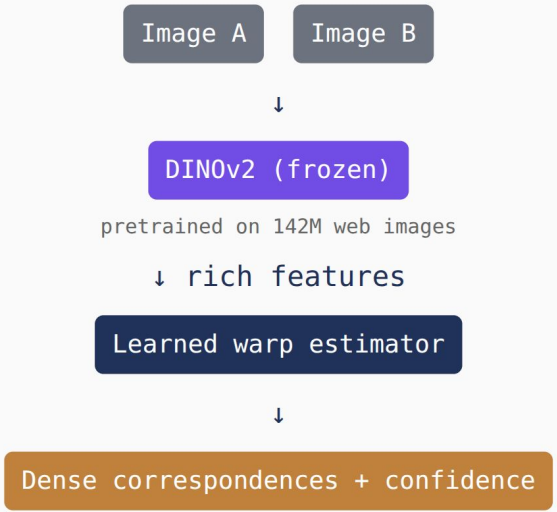
Key idea

- Models warp as a Gaussian process over learned kernel features
- Coarse-to-fine: global warp → progressive refinement
- Confidence head filters unreliable matches

*"Lowest reprojection error on harsh scenes — best for **change detection**"*

RoMa Learned Dense

Edstedt et al., 2024 · "Robust Dense Feature Matching"



Foundation-model features → **generalizes** to unseen scenes

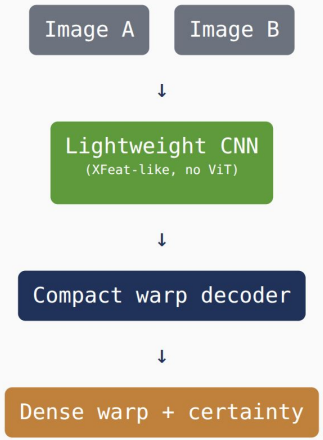
Key idea

- Foundation model (DINOv2) as frozen backbone
- Warp estimator trained on top of rich features
- No task-specific encoder training

Quality leader on harsh scenes — 170x slower than SIFT

RoMa Tiny Learned Dense USED IN OUR BENCHMARK

tiny_roma_v1_outdoor · lightweight sibling of full RoMa for practical deployment



Full RoMa: DINOv2 + VGG + transformer (~heavy)
RoMa Tiny: compact CNN only (~order-of-magnitude fewer params)

Why choose the Tiny variant

- **Fits in practical VRAM** — full RoMa needs upsample_res=560 + model reload every 10 pairs to stay within 32 GB
- **~4–5x faster** than full RoMa on the same images
- **Same warp-estimation formulation** — just with a lighter backbone
- **Designed for deployment:** ROVs / AUVs / ship-side workstations

All "RoMa" numbers in this benchmark come from the Tiny variant. Full RoMa likely beats it further on hydrothermal at 3–5x cost.

Evaluation Metrics

Reconstruction quality

- N_i — Registered images
- $N_{\square}$ — Triangulated 3D points
- $\bar{e}$ — Mean reprojection error (px)
- τ — Mean track length
- $\bar{\delta}$ — Co-visible points per image

Pragmatic

- **t** — Total pipeline time (min)
- **Inliers vs viewing angle** — viewpoint-robustness diagnostic
- **Co-visibility ratio** — scene-coverage diagnostic

We report the **largest sub-model** from COLMAP's mapper — when fragmentation occurs, it's a descriptor failure signal, not a bug to aggregate around.

Results: All Datasets

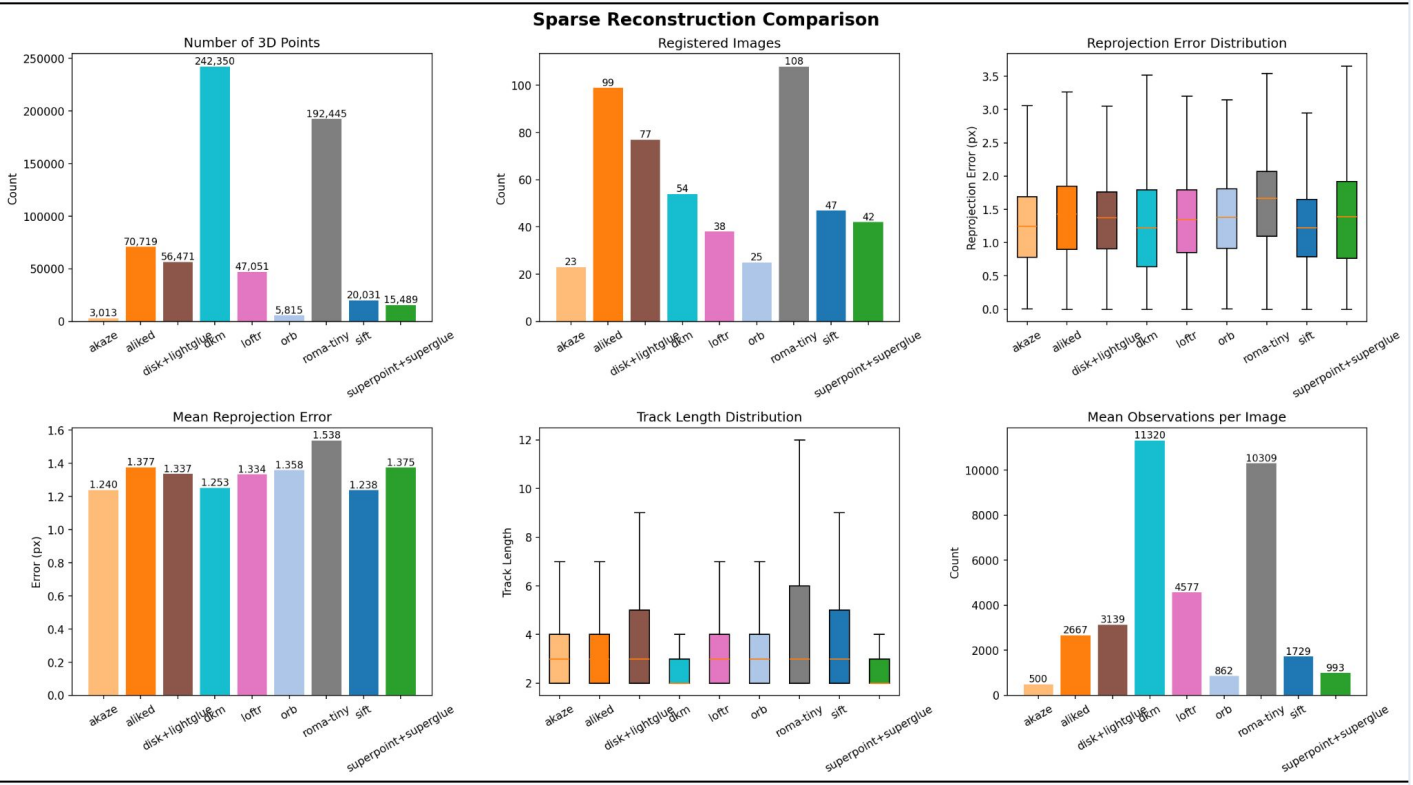
Results: All Datasets

Method	Glacier (66)				Hydrothermal (108)				UAV City Hall (65)			
	N_r	N_p	$\bar{o}$	t(m)	N_r	N_p	$\bar{o}$	t(m)	N_r	N_p	$\bar{o}$	t(m)
SIFT	**66**	70,681	5.1k	**3.3**	47	20,031	1.7k	**3.2**	**65**	63,037	4.4k	**1.8**
ORB	64	17,691	1.1k	5.0	25	5,815	0.9k	8.0	50	12,072	0.9k	3.9
AKAZE	58	12,108	0.7k	8.0	23	3,013	0.5k	3.8	52	9,703	0.7k	3.2
ALIKED	63	63,422	6.0k	10.3	99	70,719	2.7k	27.4	**65**	69,432	5.7k	8.6
SP + SG	63	37,951	1.9k	13.1	42	15,489	1.0k	21.3	**65**	45,736	2.4k	13.4
DISK + LG	63	63,621	5.0k	5.7	77	56,471	3.1k	7.6	**65**	67,323	4.8k	3.4
LoFTR	63	215,945	11.3k	61.7	38	47,051	4.6k	92.6	**65**	153,367	9.1k	—
RoMa	**66**	**222,191**	**18.8k**	152	**108**	192,445	10.3k	549	**65**	93,526	**9.5k**	197
DKM	**66**	75,120	2.6k	79	54	**242,350**	**11.3k**	174	**65**	**189,857**	7.9k	50

N_r = registered images, N_p = 3D points, $\bar{o}$ = co-visible points/image, t = minutes. Green = best per column. Red = notably poor.

N_i = registered images, **N□** = 3D points, **$\bar{\delta}$** = co-visible points/image, **τ** = track length, **t** = time. Green = best per column. Red = notably poor.

Reconstruction Comparison - HydroThermal

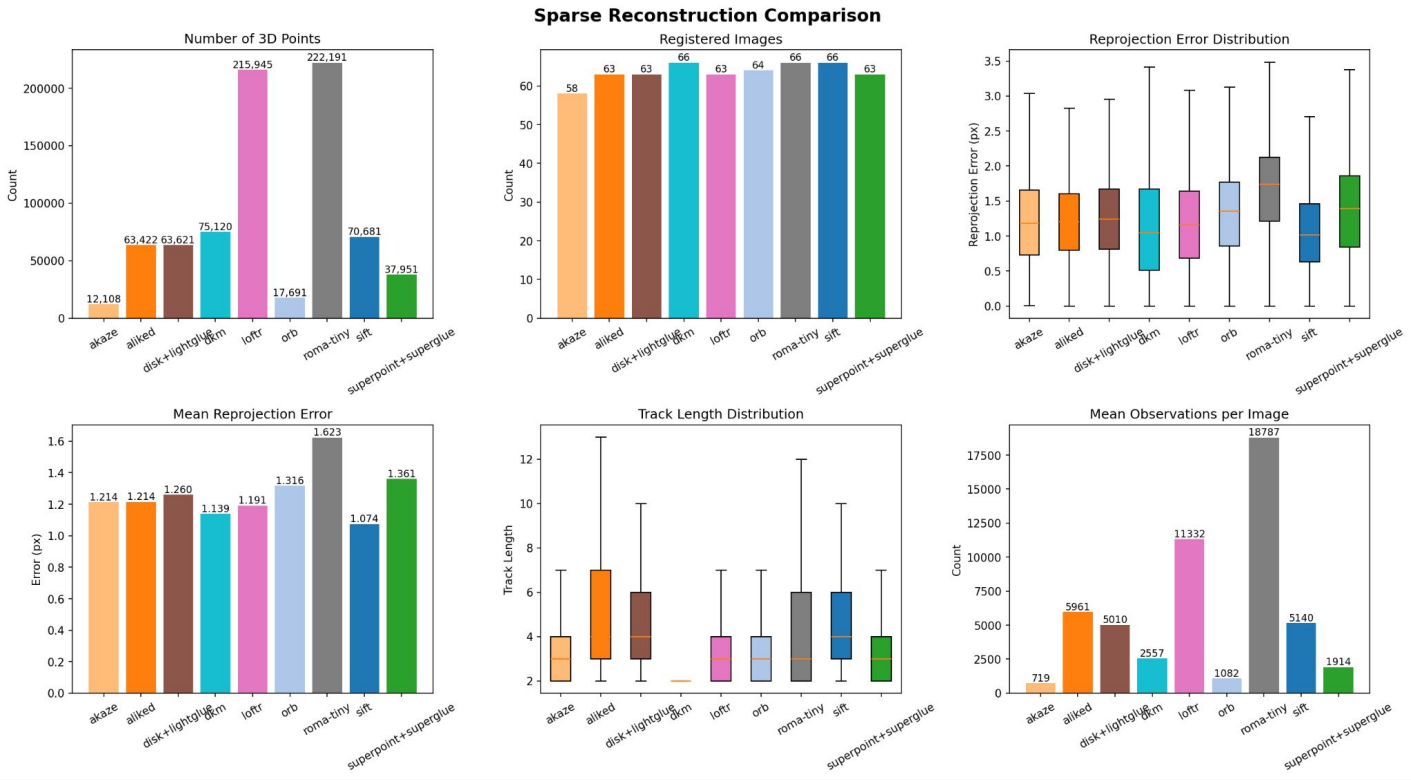


Key Takeaways

- **Only RoMa registers all 108 images**
- ALIKED closed behind at 99/108
- SIFT drops to 47/108 — partial failure
- SP+SG drops to 42/108
- DKM: only 54 imgs, but 242k points
- Binary (ORB/AKAZE): 23-25/108 △

This dataset requires **learned descriptors**.

Reconstruction Comparison - Glacier Dataset

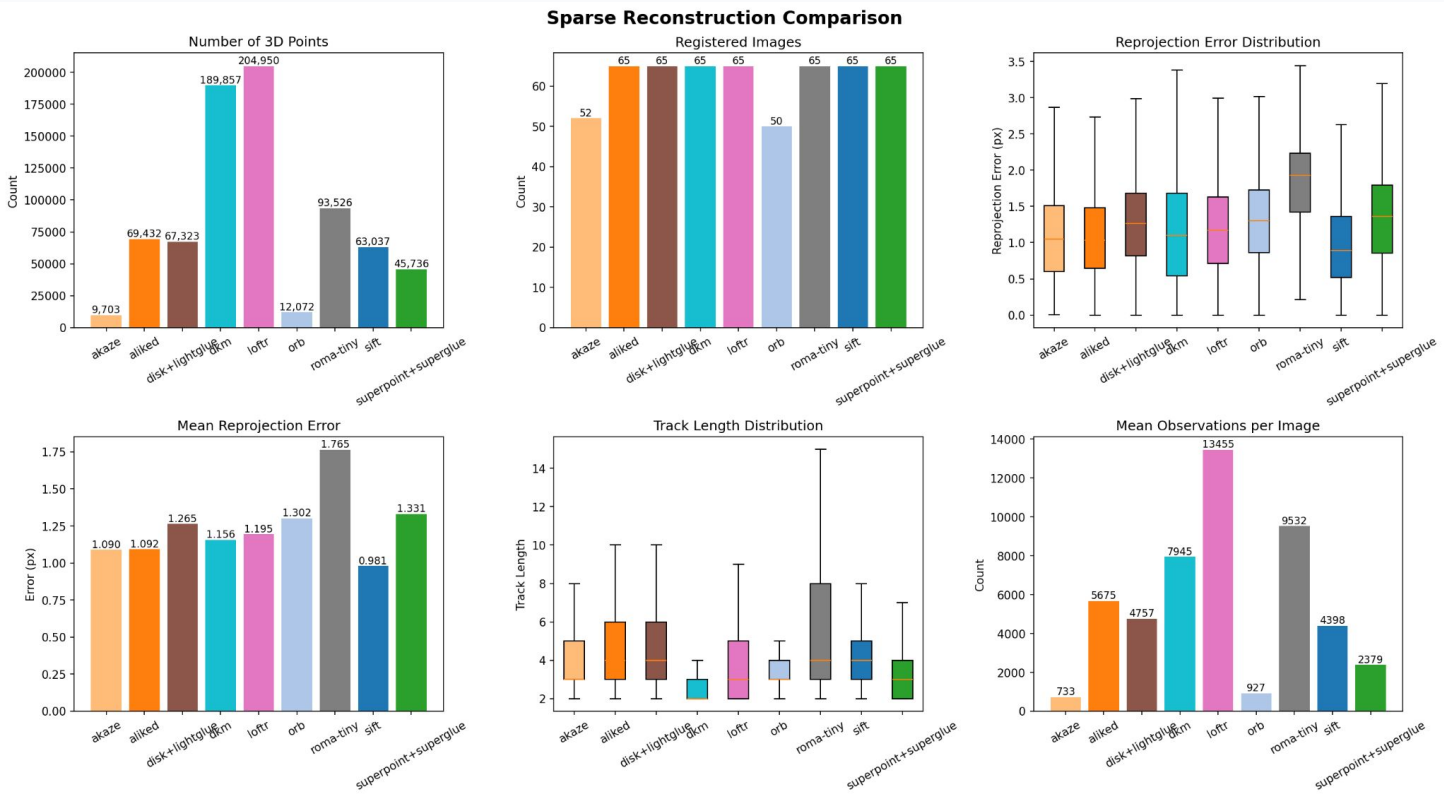


Key takeaways

- **SIFT + dense matchers** all register 66/66 ✓
- SIFT has **lowest reproj error (1.07 px)**
- Dense methods get **3-4× more** 3D points
- RoMa: 222k points, 18.8k co-visible/img
- ORB/AKAZE behind but not catastrophic

Low-texture ice degrades classical ranking but doesn't break it.

Reconstruction Comparison - CityHall Dataset

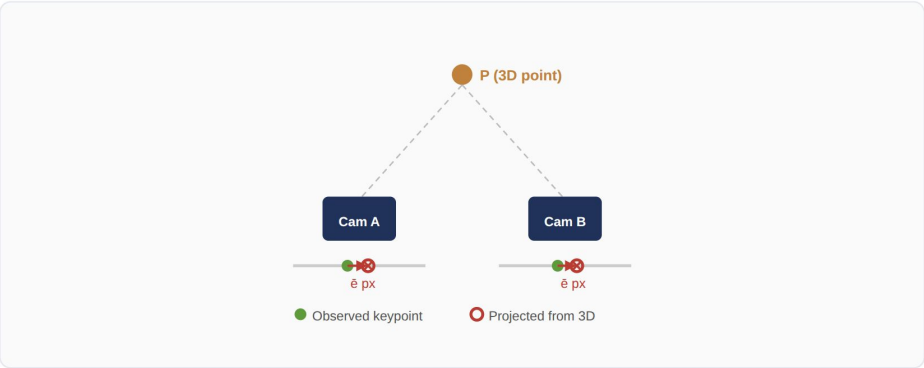


Key takeaways

- **Every non-binary method:** 65/65 ✓
- **Differentiator:** point-cloud density
- **DKM:** 190k points (winner)
- **SIFT:** 63k points but lowest error (0.98 px)
- **ORB/AKAZE:** still worst, but workable

Well-textured scene — compute cost, not capability, becomes the discriminator.

Reprojection Error - What It Means



ē = mean distance (px) between where a 3D point projects and where it was observed

Why it matters

- **Geometric accuracy** of each triangulated 3D point
- Low ē → points land where cameras actually saw them → **trustworthy geometry**
- High ē → point placement is uncertain → phantom "changes" in change detection

Our numbers

Method	Glacier	Hydro	UAV
SIFT	**1.07**	1.24	**0.98**
DKM	1.14	**1.20**	1.16
LoFTR	1.19	1.33	1.21
ALIKED	1.21	1.38	1.09
DKM	1.14	**1.20**	1.16
RoMa	1.62	1.54	1.76

⚠ RoMa registers **the most images** but has the **highest reprojection error** — it matches broadly but less precisely.

Reprojection Error - Which Method to Choose

Low ē = Precise geometry

Critical when you compare 3D structure *across time*:

Change detection / repeat surveys

An uncertainty of 0.5 px extra in ē can displace a triangulated point by centimetres at typical ROV working range — showing as false change.

Photogrammetric measurement

Mapping coral growth, glacier retreat — all need sub-pixel geometric fidelity.

Decision guide based on ē

Scene	Task	Best method	ē (px)
Well-textured	Any	**SIFT**	**0.98–1.07**
Glacier / moderate	Change detect.	**DKM**	**1.14**
Glacier / moderate	Coverage	**RoMa Tiny**	1.62
Turbid / deep-sea	Change detect.	**DKM**	**1.20**
Turbid / deep-sea	Map coverage	**RoMa Tiny**	1.54

Key insight: DKM achieves near-SIFT precision on harsh scenes that SIFT can't even reconstruct. This makes DKM the best of both worlds for **geometry-sensitive** OCEANS tasks.

Inliers vs Co-visible Points

Two related but **different metrics** — both are important for understanding matcher quality.

True Inliers

Matches that **pass F-matrix RANSAC** (from `two_view_geometries`).

- Captured after matching
- Per-pair count
- Measures matcher quality

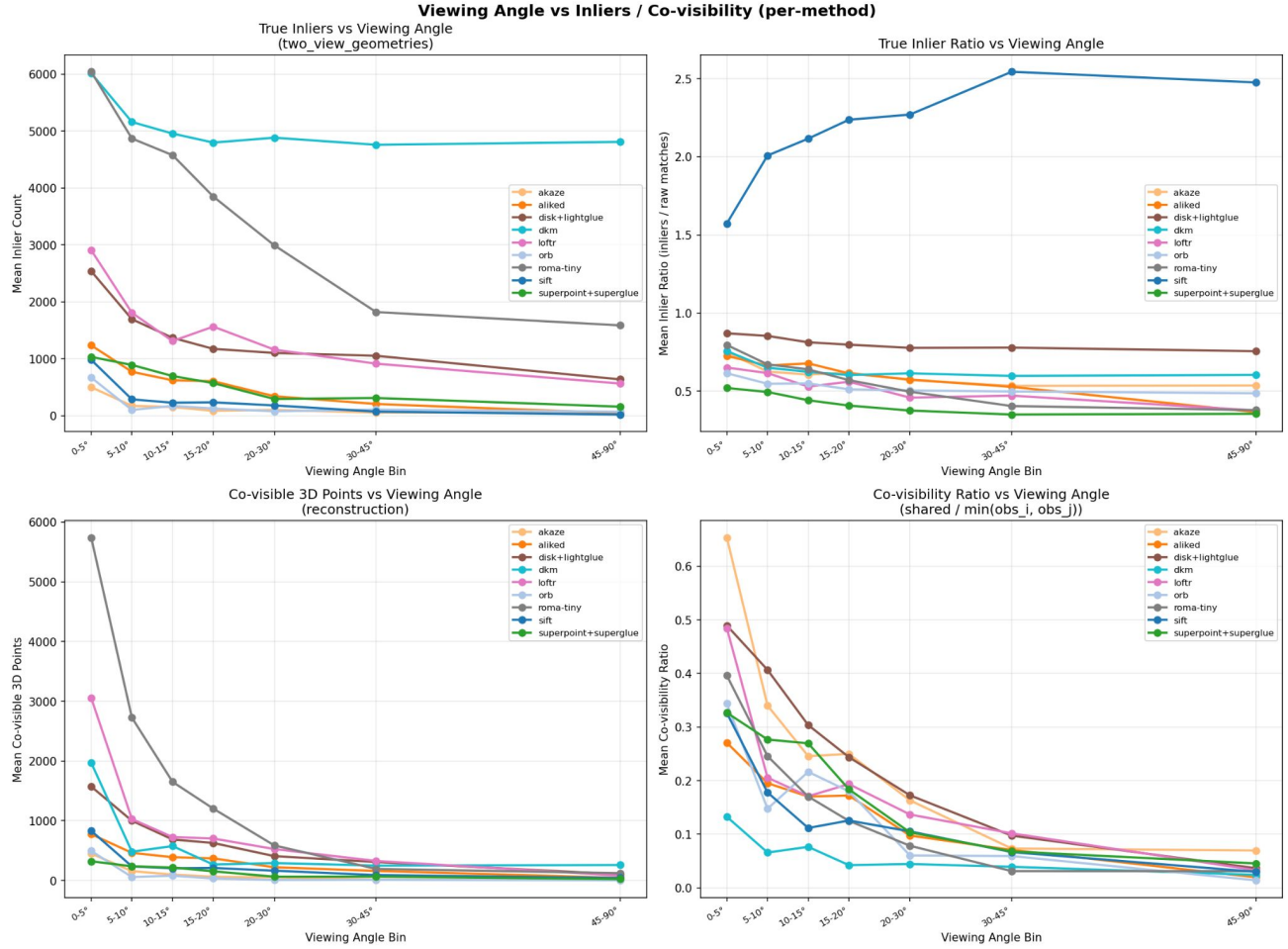
Co-visible 3D Points

3D points triangulated in the final map that **both images observe**.

- Captured after mapper
- Per-pair or per-image average
- Measures reconstruction contribution

Gap between them = matches that survived RANSAC but failed triangulation (small baseline, degenerate geometry).

Viewpoint Robustness — Hydrothermal

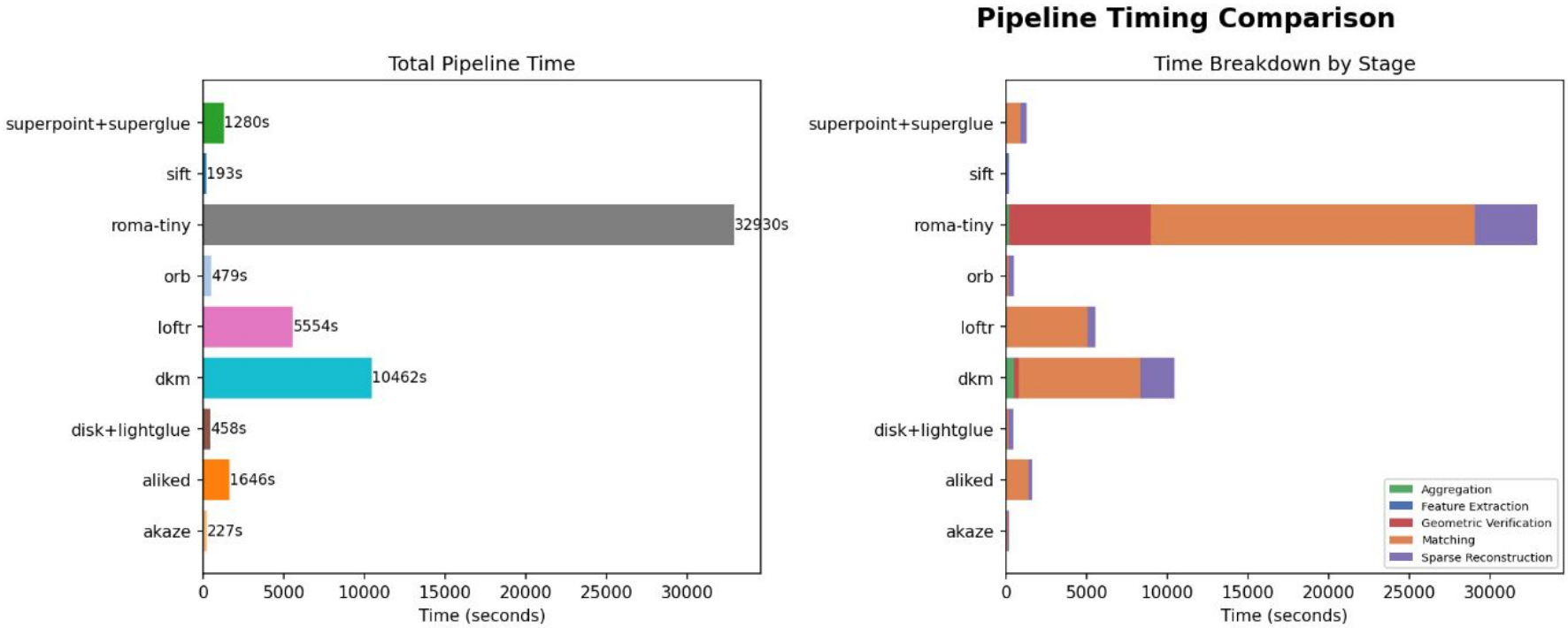


How to read

X-axis: angle bin between camera pair
Y-axis: mean inliers / co-visibility

- **Dense matchers** retain inliers past 20°
- **Sparse** drops after 10–15°
- **ORB/AKAZE:** near-flatline

Timing — Hydrothermal Vent



Data Source: Comparative timing results based on standardized Hydrothermal Vent benchmark sequences.

Timing — Hydrothermal Vent

Method	Time
SIFT	**3.2 min**
AKAZE	3.8 min
DISK+LG	7.6 min
ORB	8.0 min
SP+SG	21.3 min
ALIKED	27.4 min
LoFTR	92.6 min
DKM	174 min
RoMa	**549 min (9.1 h)**

Headline Analysis

RoMa demonstrates significantly higher computational intensity compared to traditional feature-based methods.

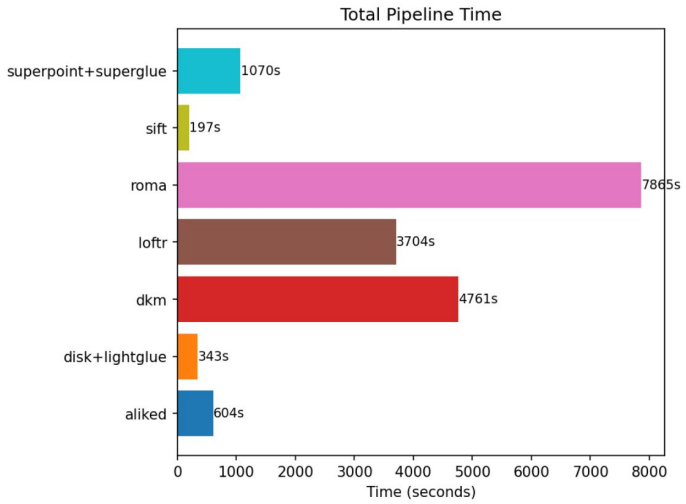
RoMa is 170× slower than SIFT
DKM is 54× slower than SIFT

This represents the primary headline compute cost on the Hydrothermal dataset.

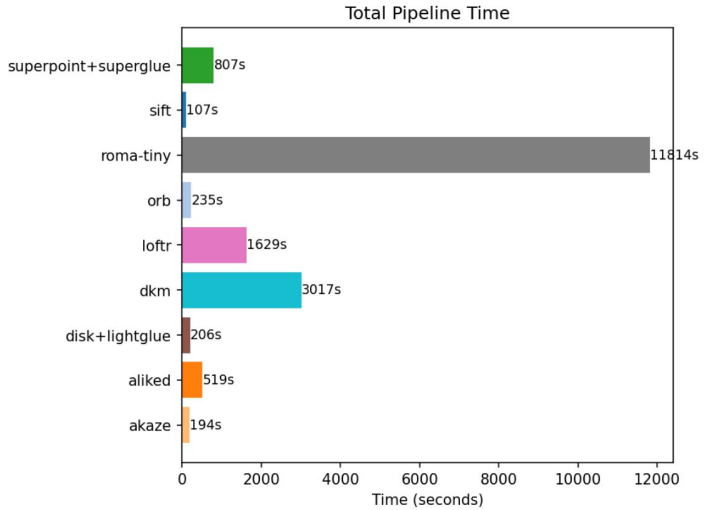
Data Source: Comparative timing results based on standardized Hydrothermal Vent benchmark sequences.

Timing — Glacier & UAV

Glacier



UAV City Hall



Scale of each bar reflects total pipeline time. Dense methods consistently dominate costs.

Co-visibility Insight: RoMa vs DKM

Hydrothermal: DKM has **more 3D points (242k)** but similar per-image co-visibility (δ=11.3k) to RoMa (10.3k), despite registering only half the images.

RoMa — spread

Might spreads points **across the full sequence**. 108/108 cameras all participate.

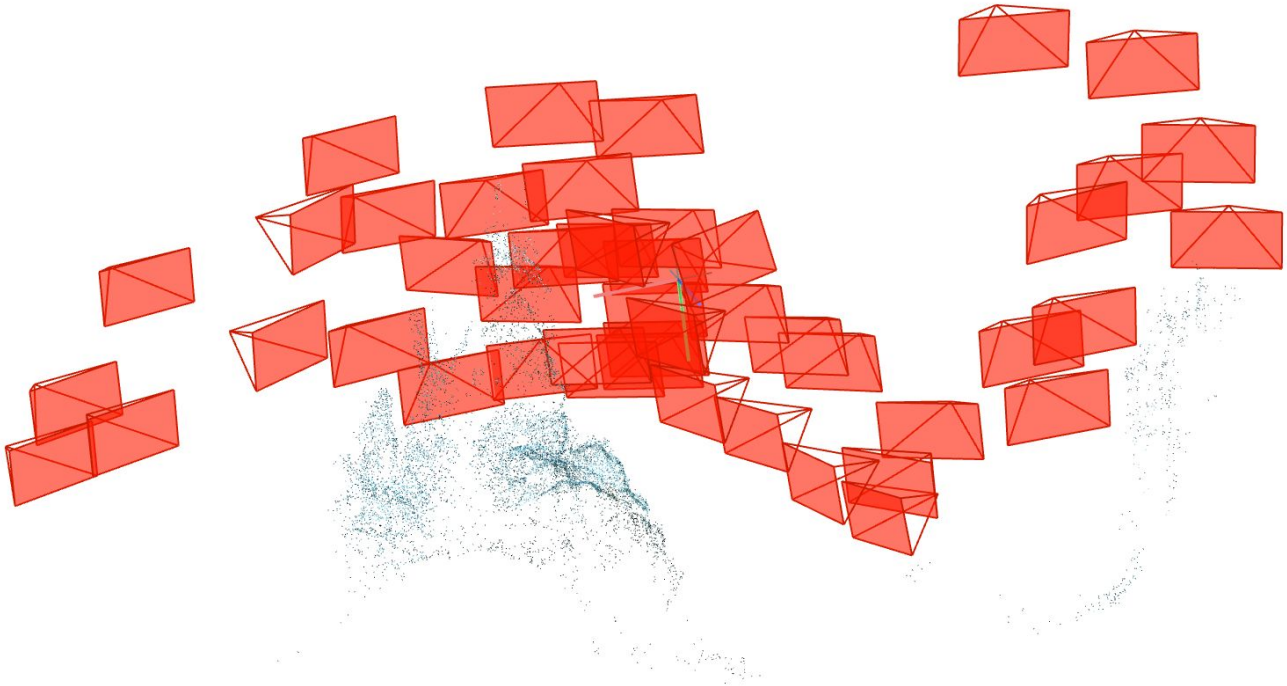
Question:

- **Still Covisible points should be better, as 2x images have been registered.**

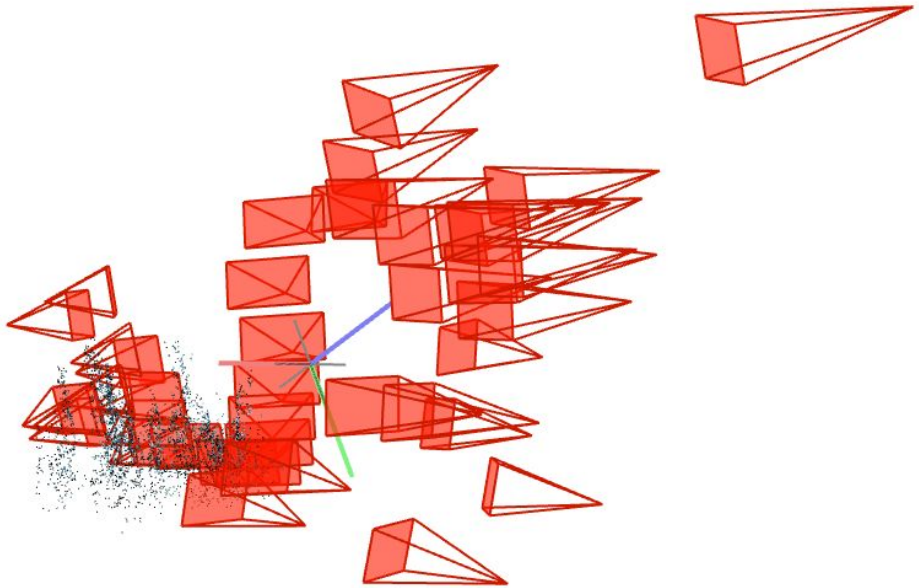
DKM — density

Might concentrates points in a **smaller dense model**. Only 54 cameras, but very dense locally.

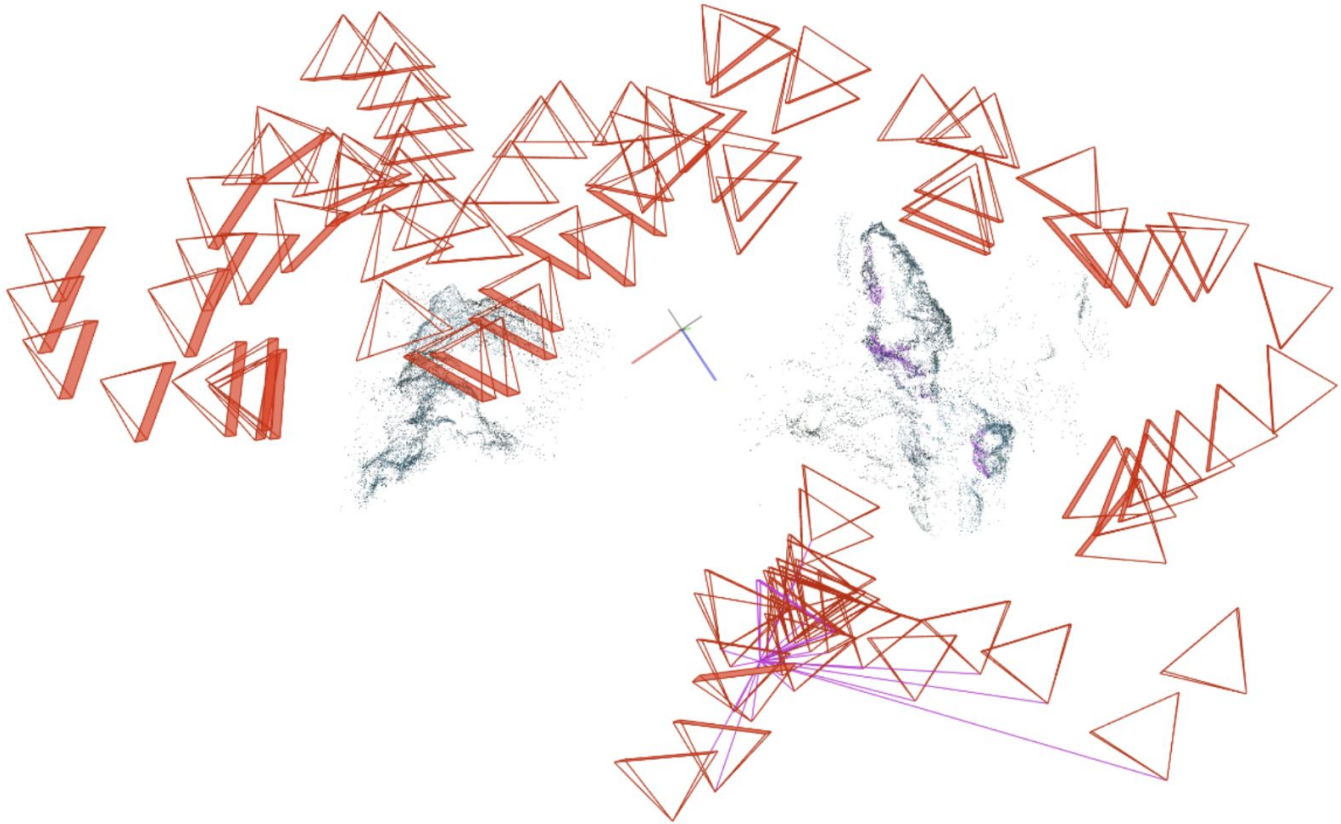
SIFT - Sparse Reconstruction - HydroThermal



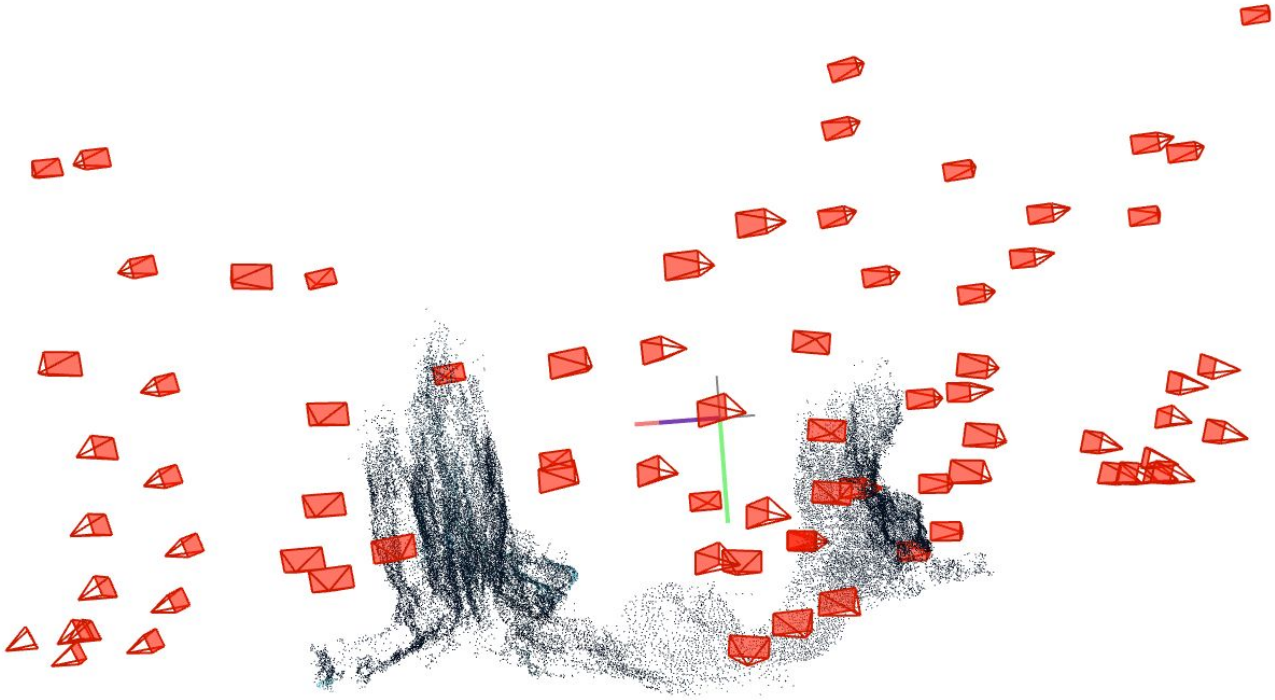
Superpoint + SuperGlue - Sparse Reconstruction



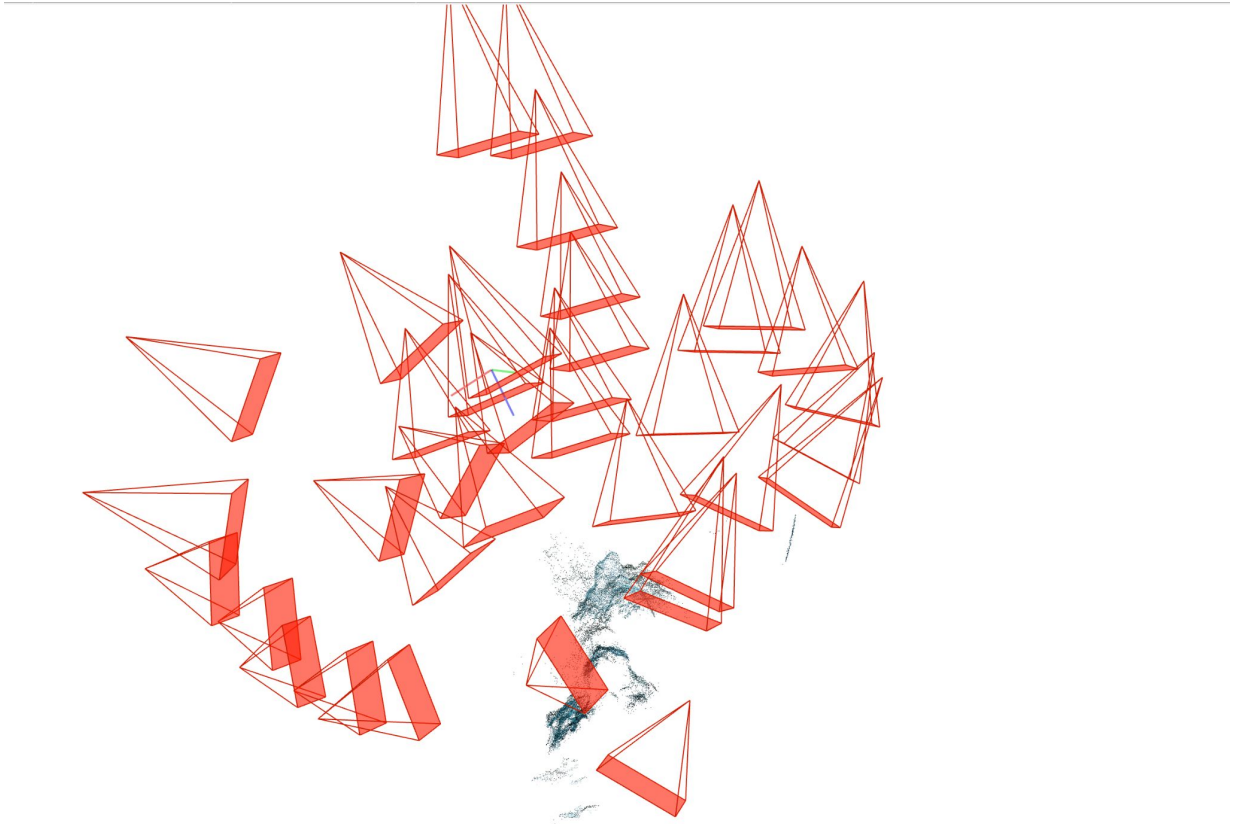
Aliked - Sparse Reconstruction - Hydrothermal



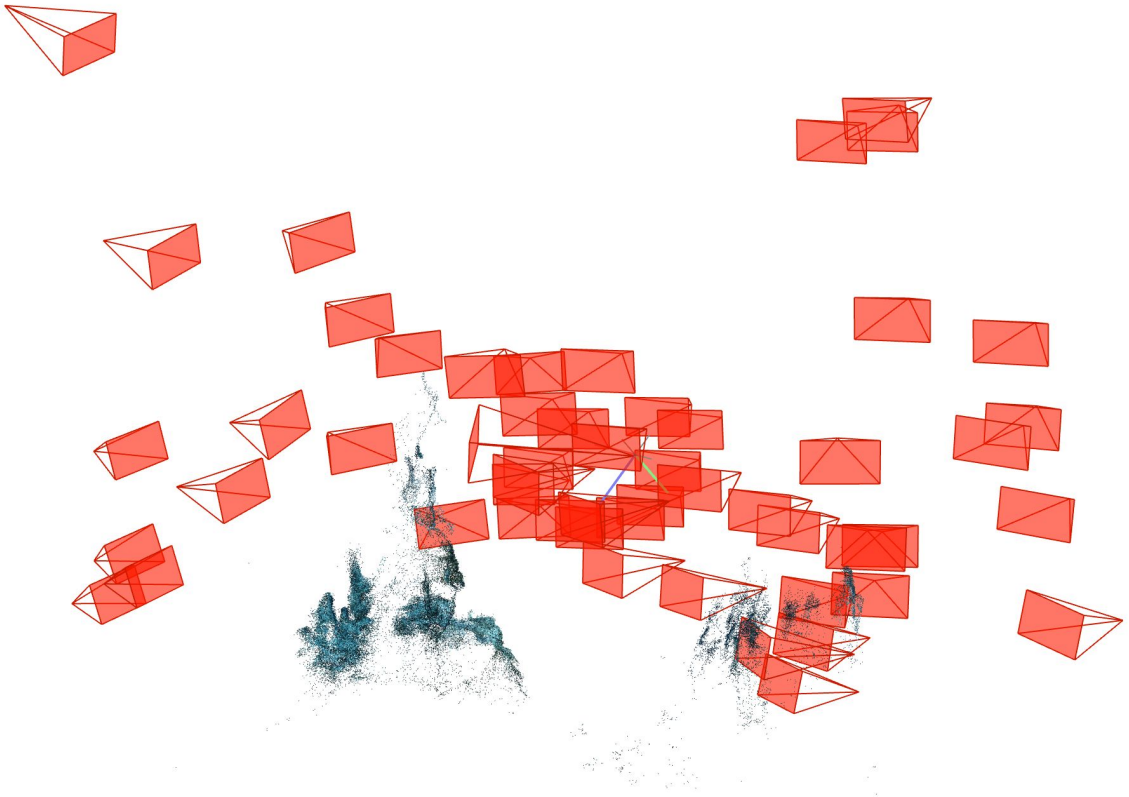
DISK + LightGlue - Sparse Reconstruction -HydroThermal



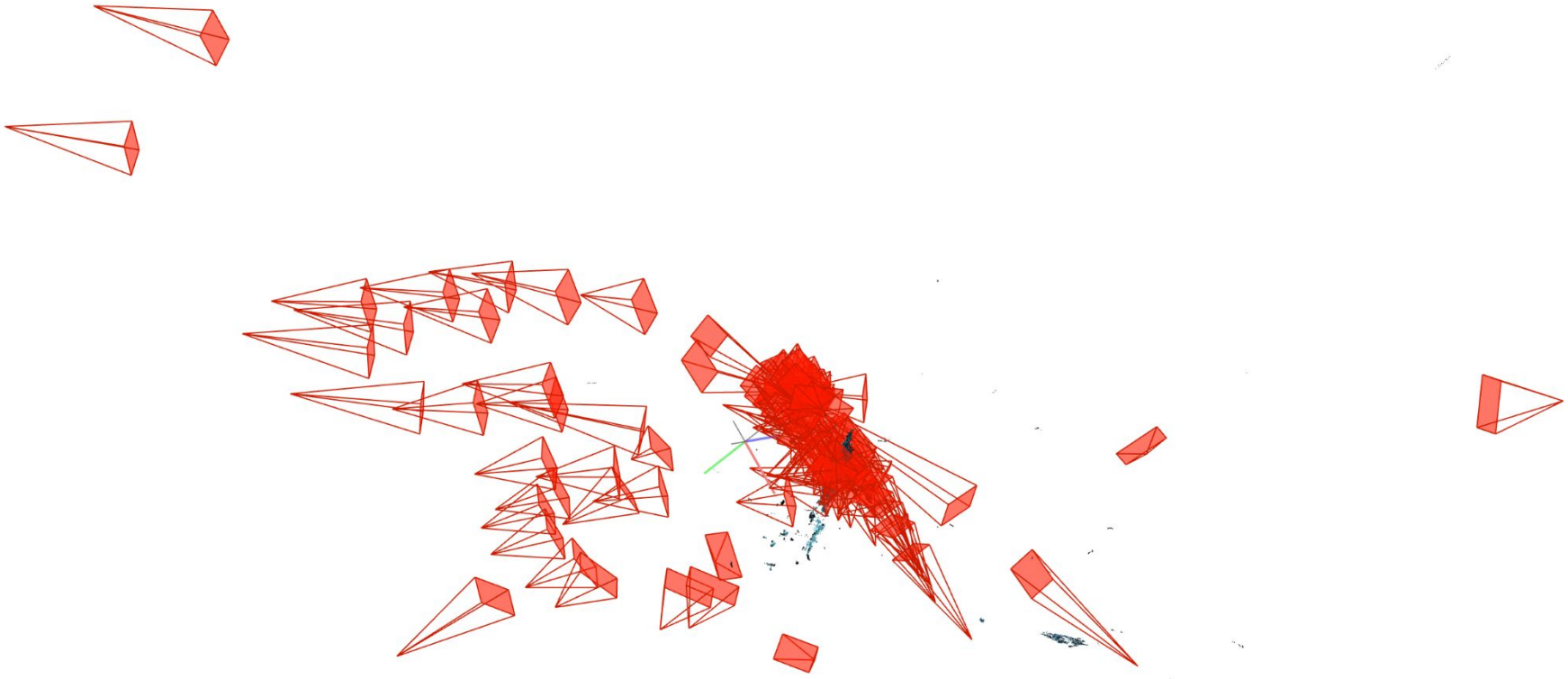
LoFtr - Sparse Reconstruction



DKM - Sparse Reconstruction - HydroThermal



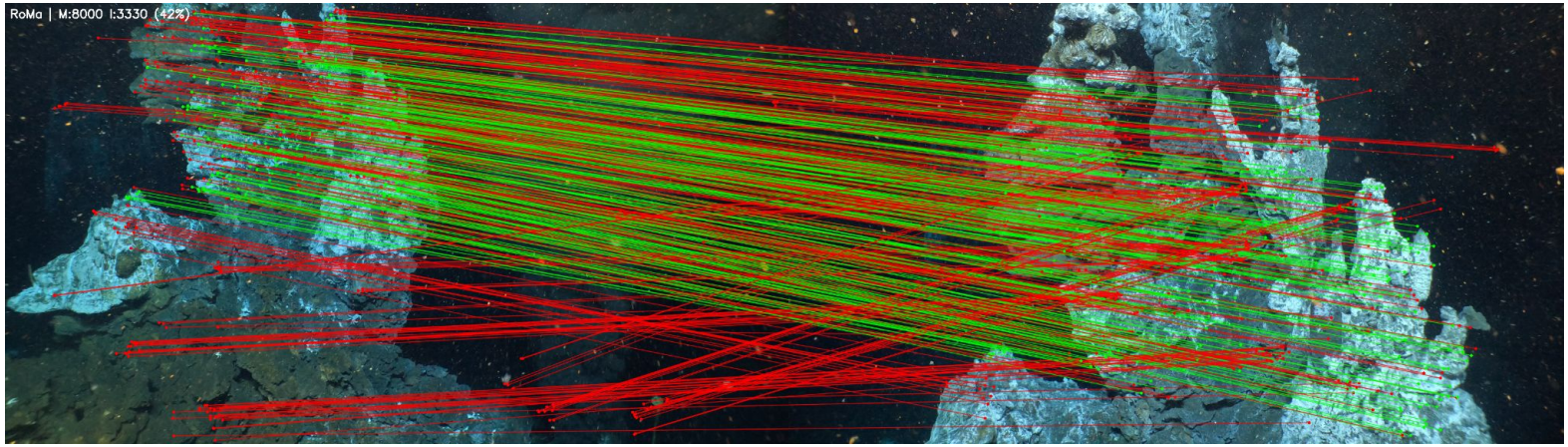
RoMA-Tiny - Sparse Reconstruction

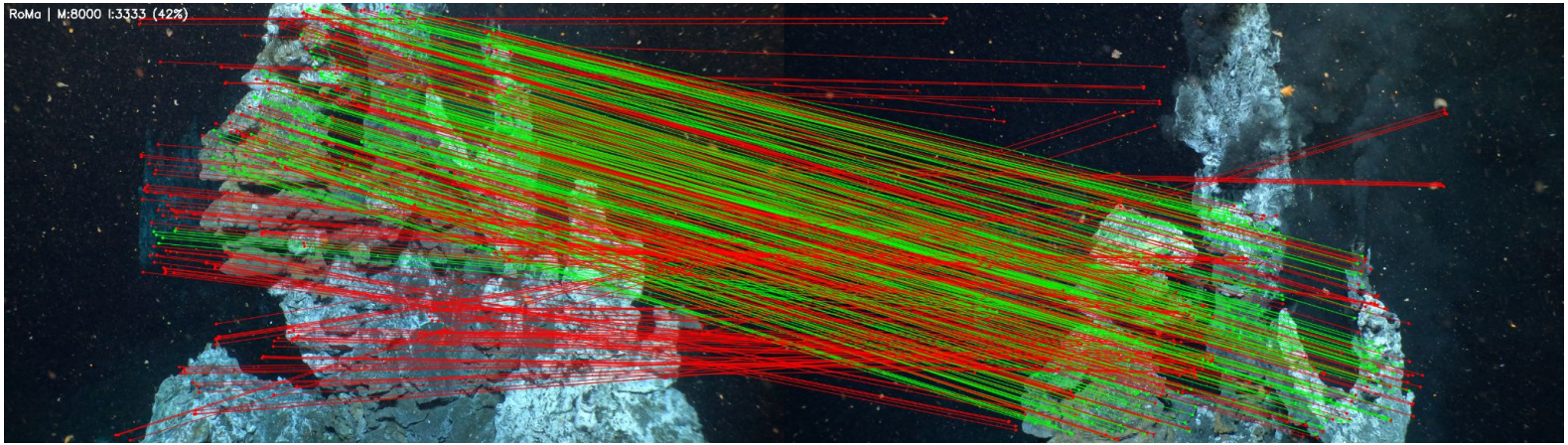


RoMa Tiny - Sparse Reconstruction - Glacier Dataset (26 Aug)

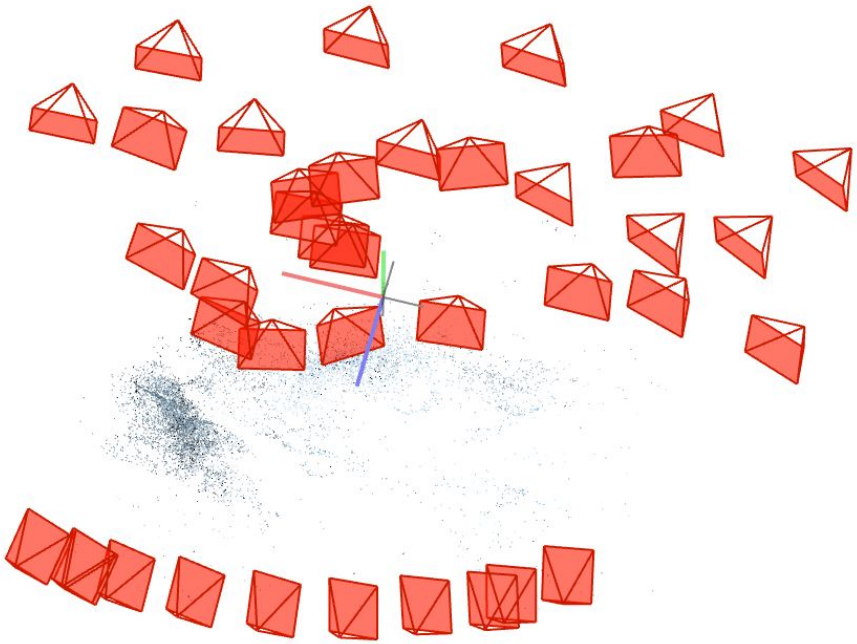


RoMa (Tiny) Matches Comparison





RoMa Full - Sparse Reconstruction - Glacier Dataset (25 Aug)



Why RoMa Does’nt Perform good ?

1. Dense Matcher (We aggregate keypoints)
2. RoMa Tiny suffers 15-20% degradation compared to Roma Full.
   a. Reported in GitHub
3. Aggregation Pixels be dataset and model dependant? (No Aggregation -> Noise) (More Aggregation Inaccurate)
4. Aggregation might help them pass epipolar line test
5. RANSAC finds a wrong but consistent model
6. Lower RANSAC threshold - Can work for Roma and DKM, but not for Roma-tiny, which gives poor quality matches

Combined Results & Comparison

CPU: 13th Gen Intel(R) Core(TM) i9-13900K

GPU: NVIDIA RTX 5090

TABLE II

RECONSTRUCTION QUALITY AND RUNTIME ACROSS DATASETS. BOLD = BEST PER COLUMN (FOR $\bar{e}$ AND t: LOWEST). N_r = REGISTERED IMAGES, N_p = TRIANGULATED 3D POINTS, $\bar{e}$ = MEAN REPROJECTION ERROR (PX) — THE DIRECT INDICATOR OF TRIANGULATION PRECISION AND THUS CHANGE-DETECTION SUITABILITY, $\bar{o}$ = MEAN CO-VISIBLE 3D POINTS PER REGISTERED IMAGE, t = TOTAL PIPELINE TIME IN MINUTES. TOTALS: GLACIER 66 IMAGES, HYDROTHERMAL 108, UAV 65. RUNTIME ON A SINGLE NVIDIA GPU; — INDICATES NOT RUN.

	Glacier (aerial)					**Hydrothermal** (underwater)					**UAV City Hall**				
Method	N_r	N_p	$\bar{e}$	$\bar{o}$	t (m)	N_r	N_p	$\bar{e}$	$\bar{o}$	t (m)	N_r	N_p	$\bar{e}$	$\bar{o}$	t (m)
SIFT	**66**	70,681	**1.07**	5.1k	**3.3**	47	20,031	1.24	1.7k	**3.2**	**65**	63,037	**0.98**	4.4k	**1.8**
ORB	64	17,691	1.32	1.1k	5.0	25	5,815	1.36	0.9k	8.0	50	12,072	1.30	0.9k	3.9
AKAZE	58	12,108	1.21	0.7k	8.0	23	3,013	1.24	0.5k	3.8	52	9,703	1.09	0.7k	3.2
ALIKED	63	63,422	1.21	6.0k	10.3	99	70,719	1.38	2.7k	27.4	**65**	69,432	1.09	5.7k	8.6
SP + SG	63	37,951	1.36	1.9k	13.1	42	15,489	1.37	1.0k	21.3	**65**	45,736	1.33	2.4k	13.4
DISK + LG	63	63,621	1.26	5.0k	5.7	77	56,471	1.34	3.1k	7.6	**65**	67,323	1.26	4.8k	3.4
LoFTR	63	215,945	1.19	11.3k	61.7	38	47,051	1.33	4.6k	92.6	**65**	153,367	1.21	9.1k	27.15
RoMa	**66**	**222,191**	1.62	**18.8k**	152	**108**	192,445	1.54	10.3k	549	**65**	93,526	1.76	**9.5k**	197
DKM	**66**	75,120	1.14	2.6k	79	54	**242,350**	**1.20**	**11.3k**	174	**65**	**189,857**	1.16	7.9k	50

① Can dense matchers be used for harsh scenes ????

Performance on hydrothermal vent:

- RoMa: 108/108 images - Reconstruction is Garbage
- ALIKED: 99/108 - Decent Reconstruction
- DISK+LG: 77/108 - Better Reconstruction than Aliked
- DKM: 54/108 - Good Reconstruction
- SIFT: 47/108 - Okayish
- SP+SG: 42/108 - Okayish
- ORB/AKAZE: 23-25/108

DKM is the only Dense Matcher that does good job - Roma registers all images, but suffers from really high reprojection error and gives a garbage reconstruction

Registration success ≠ correct geometry

② ML Based Matchers comes at substantial compute cost

170x

RoMa vs SIFT on hydrothermal

9.1 hours vs 3.2 minutes

54.4x

DKM vs SIFT on hydrothermal

174 minutes vs 3.2 minutes

~9x

Aliked vs SIFT on hydrothermal

9 minutes vs 3.2 minutes

- **DKM** = 3× faster than RoMa (174 min)
- **LoFTR** = 2× faster than DKM (93 min)
- **ALIKED** = ~10x Faster than LoFTR (27 min)

This matters for **onboard processing** and **ship-side post-mission analysis**.

③ Sparse learned = best quality/compute trade-off

ALIKED

Hydrothermal: 99/108 images

27 min — 20× faster than RoMa, 6x than DKM

- Close to dense quality, without the cost

DISK + LightGlue

Hydrothermal: 77/108 images

7.6 min — faster than SP+SG

- Solid baseline for GPU-constrained deployments

For AUV/ROV payloads with tight GPU budgets: **ALIKED** or **DISK+LightGlue**.

④ SIFT is not dead

Performance Metrics

Full Registration:

- Glacier (**66/66**) and UAV (**65/65**)

Efficiency:

- **Under 4 min** on every dataset

Lowest Reprojection Error

0.98 px
on UAV dataset

1.07 px
on Glacier dataset

Where scene texture is usable, **SIFT remains a valid, fast, interpretable baseline**. Its only failure mode is photometric breakdown (turbidity, severe low-texture).

⑤ Binary descriptors are consistently weakest

Performance Breakdown

ORB (even tuned WTA_K=3, cross-check): underperforms across all 3 datasets

AKAZE: nonlinear scale space doesn't rescue it

Co-visibility: 3–10× lower than any learned method

No runtime advantage over SIFT

Their continued use in legacy marine toolchains is **not justified**. Drop them.

Trade-off: Sparse vs Dense

Sparse Methods

Compute: Fast (min)

VRAM: Low

Point Cloud: Moderate density

Integration: Native (keypoints + descs)

Weakness: Struggles with low-texture and wide baselines.

Dense Methods

Compute: Slow (hrs)

VRAM: High (RoMa reloads model)

Point Cloud: Very high density

Integration: Needs aggregation step

Strength: Shines in low-texture; degrades gracefully.

If texture is usable and time is tight → **sparse**. If scene is hard or compute is abundant → **dense**.

Trade-off: Sparse vs Dense

MAIN REASONS

1. Lower geometric precision: Dense correspondences are abundant but less accurately localized than sparse keypoints. Small pixel errors translate to large 3D triangulation errors.

2. Higher reprojection error: SfM is sensitive to modest image-space noise, making dense bundles less stable and prone to warped reconstructions.

3. Weak multi-view track consistency: Dense matchers are optimized for pairwise correspondences, not for building long, repeatable feature tracks across many views.

4. Soft error distribution: Instead of a few obvious outliers, dense matchers often produce many slightly wrong matches, which are harder for robust estimators like RANSAC to reject.

DENSE VS. SPARSE FOR SFM

Dense matchers

- Very high match count
- Better scene coverage
- Often noisier localization
- Pairwise success ≠ stable geometry

Sparse matchers

- Fewer but sharper keypoints
- Lower reprojection error
- Longer, more stable tracks
- Usually better for BA and SfM

Core Insight
SfM is a **precision** problem, not just recall.

Best Practice
Use dense matches **only after filtering and sparsification.**

One-Line Takeaway

Dense matchers register **more images**, but their noisier and less consistent correspondences often lead to **worse global reconstruction quality.**

Trade-off: Geometric vs Learned

Geometric (SIFT, ORB...)

Generalization: Handcrafted, predictable

Repeatability: Excellent (SIFT)

Texture/Turbidity: Moderate on low/ice

Compute: Very fast on CPU

Interpretability: Well understood

Learned (DISK, DKM...)

Generalization: Domain gap possible

Repeatability: Competitive

Texture/Turbidity: Major improvement

Compute: GPU required

Interpretability: Black box

Binary descriptors (ORB, AKAZE) are strictly dominated — no compute advantage over SIFT, and much worse quality.

Deployment Recommendations

Scene	Compute	Recommendation
Well-textured buildings, rocks, coast	Any	**SIFT** fast, accurate
Mid-difficulty glacier, moderate biology	GPU available	**ALIKED or DISK+LG**
Low-texture turbid, biological clutter	Ample compute	**DKM** (quality)
Low-texture dense cloud wanted	Moderate compute	**DKM** (fastest dense)
Legacy ORB/AKAZE pipeline	Any	**Migrate** no advantage

Applications + Key Findings

Applications

Photogrammetric 3D reconstruction underpins:

- Seafloor mapping
- Vent community ecology
- Glacier mass balance
- Coastal structural assessment
- Heritage documentation

Key Findings & Observations

Operators often pick descriptors **by default**.

On hydrothermal data: SIFT registers 47/108. RoMa registers 108/108.

The data isn't unusable — **the descriptor was.**

This benchmark gives OCEANS practitioners a concrete condition — descriptor mapping, on data that **resembles real missions** rather than indoor rooms.

Conclusion & Deployment Guide

Optimal descriptor choice depends on **scene difficulty, compute budget**, and **downstream task**.

Scene Condition	Compute Context	Recommended Pipeline
Well-textured buildings, rocks, speed critical	Any	**SIFT** fast, accurate, interpretable
Mid-difficulty glacier, biological clutter	GPU available	**ALIKED or DISK+LG** ~20x faster than dense
Turbid / Ice goal = scene coverage	Ample compute	**RoMa** highest registration rate
Turbid / Ice goal = change detection	Moderate compute	**DKM** most accurate dense points
Legacy Systems ORB / AKAZE pipelines	Any	**Migrate** no quality/speed advantage

Future Work

Method Axis

- **Fine-tuning** learned matchers on in-domain data (DINOv2 adapters, LoRA on RoMa)
- **Underwater-specific preprocessing** (color restoration, turbidity removal) before matching - Khubaib
- **Hybrid pipelines:** SIFT for quality on textured regions + dense matcher for low-texture patches
- **Quantized RoMa / DKM** for onboard AUV/ROV deployment

Benchmark Axis

- **Expanded to more datasets:** coral reefs, shipwrecks, sea-ice, polar imagery
- **Temporal / change-detection benchmark:** pairs of repeat surveys with ground-truth change masks
- **Power + thermal budget** on embedded GPUs (Jetson, Orin)
- **Loop-closure quality** in SLAM as a downstream evaluation
- **Robustness to platform motion blur**

Code, configurations, and COLMAP databases are present — **extensible to new datasets with a single script.**

Chapter 2: Change Detection

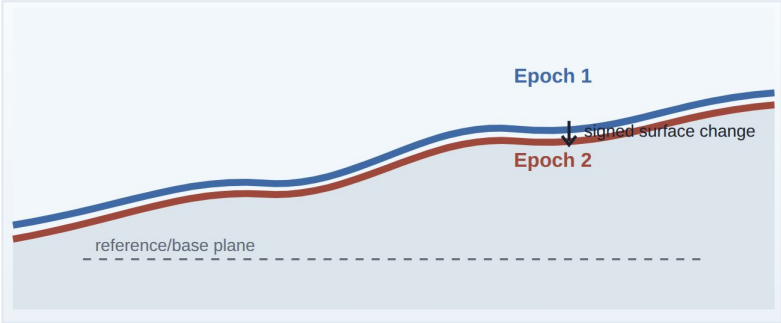
Volume Change Estimation:

1. Volume Change using Subset of Images from both Flights
2. Joint Reconstruction (Reconstruct both flights in 1 reconstruction)
3. Separate Reconstruction (Align 2 separately reconstructions)

The Problem

Challenge:

1. Glaciers might change during mid-flight
2. Scale Issues in separate reconstructions
3. Change Detection Using Subset of Images



Quantifying change through **joint reconstruction** and **automated GPS/Feature alignment** for precise glacier monitoring.

Why this is a difficult problem

Field And Glacier Effects

- The glacier surface can change
  - During flights: melt, snowfall, slush, wind redistribution, shadows, and water flow can alter the visible surface.
- Lighting can change mid-flight, which changes feature contrast on snow and ice even if geometry did not change.
- Snow and ice are low-texture, reflective, and repetitive, so feature matching can lock onto plausible but wrong areas.
- Drone flight altitude, camera angle, and image overlap may differ between epochs, producing different point density and coverage.

Reconstruction And Scale Effects

- Separate COLMAP reconstructions have arbitrary scale unless constrained by GPS, GCPs, or known distances.
- Even with GPS, horizontal scale can be reasonable while vertical alignment remains biased by meters.
- Small rotations or tilts create large volume errors over thousands of square meters.
- ICP/GICP can accidentally align the changed glacier surface itself, removing real change from the final measurement.

A vertical bias of only **0.25 m** over a **9,000 m²** footprint creates about **2,250 m³** of apparent volume change. That is why stable-region anchoring, GPS-scale diagnostics, and careful volume definitions are necessary.

We are not just subtracting two point clouds

Failure Modes

- Independent reconstructions have arbitrary scale, rotation, translation, and position drift.
- Drone GPS vertical accuracy is much worse than horizontal accuracy.
- Dense cloud sampling is uneven, so raw nearest-neighbor area weighting is biased.
- Real glacier change can pull ICP/GICP if the full surface is used for alignment.
- Absolute bed/alter/before volume needs a glacier bed or reference surface, not only a surface cloud.

What The Scripts Try To Protect

- Remove non-finite points before spatial queries.
- Detect stable regions using distance, normal consistency, and DBSCAN clustering.
- Use robust GPS RANSAC / TEASER++ where correspondences can be noisy.
- Downsample to consistent voxel grids before M3C2 and volume integration.
- Report assumptions and reject implausible DEM cells or far M3C2 correspondences.

Sign convention: signed change is epoch2 - epoch1. Negative volume means surface lowering / ice loss.

SvalBard - Two Flights 1 day apart

August 25th, 2024
39 Images

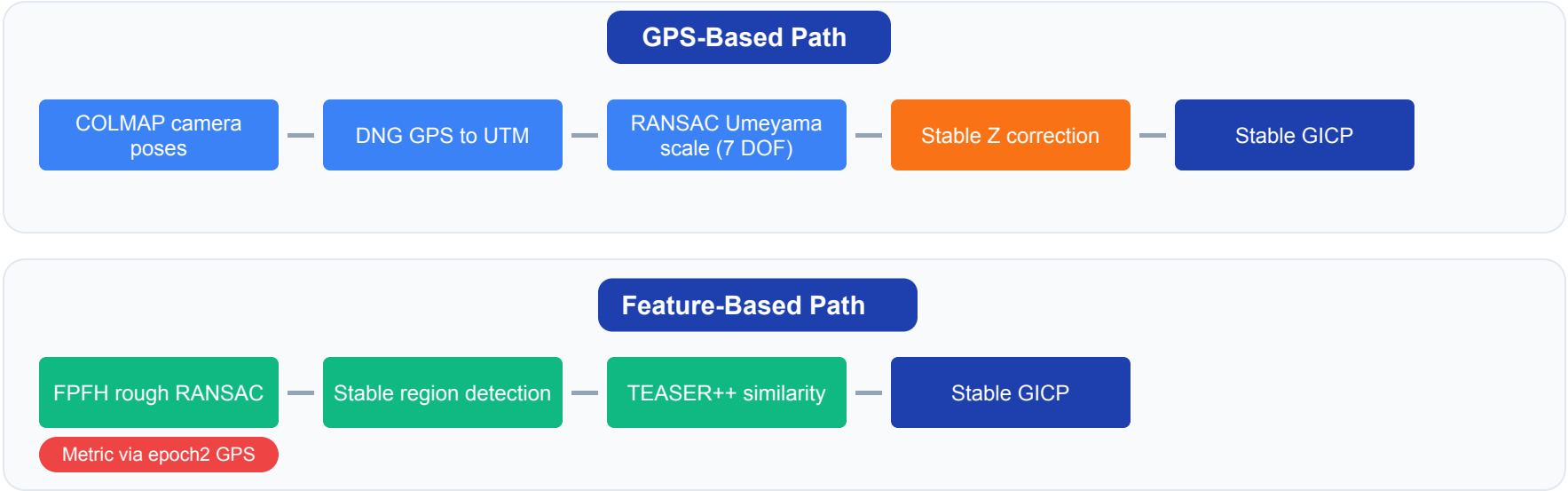


August 26th, 2024
66 Images



GPS vs. Feature Based Alignment

This script compares two ways to align independently reconstructed dense clouds: GPS-based georeferencing and feature-based registration.



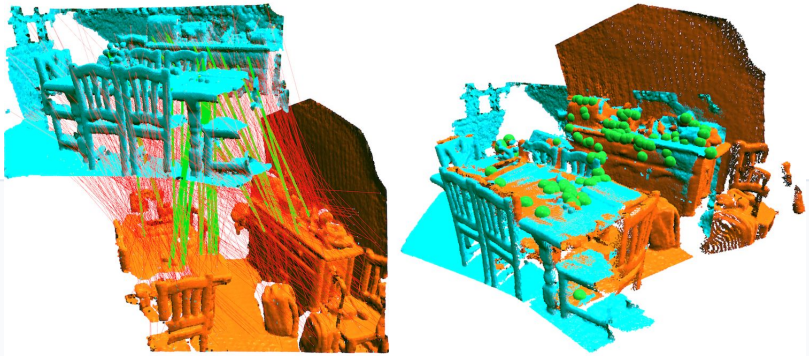
TEASER++ vs. RANSAC Comparison

RANSAC

- Random sampling of minimal sets
- Iteratively fits model + checks inliers
- **Sensitive to outlier ratio**
- Performance ↓ when outliers ↑
- Probabilistic (no strict guarantees)
- Widely used in SfM / SLAM

TEASER++

- Certifiable robust registration
- Decouples scale, rotation, translation
- **Handles extreme outliers (>99%)**
- Uses truncated least squares + graph theory
- Deterministic + globally optimal
- Strong for harsh / mismatched correspondences



Correspondences

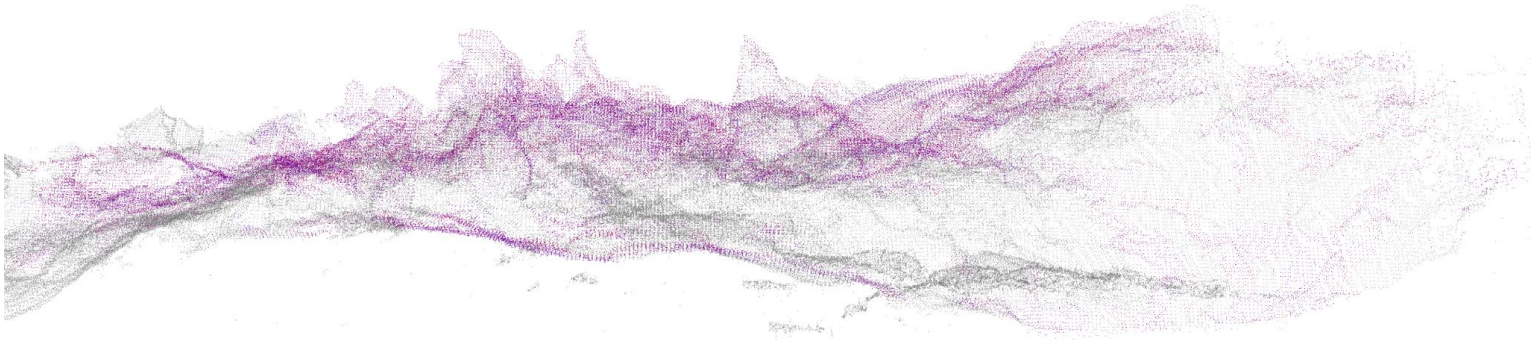
TEASER++ Alignment

Key Insight: RANSAC is fast & simple, whereas **TEASER++** is robust under extreme noise and outlier conditions.

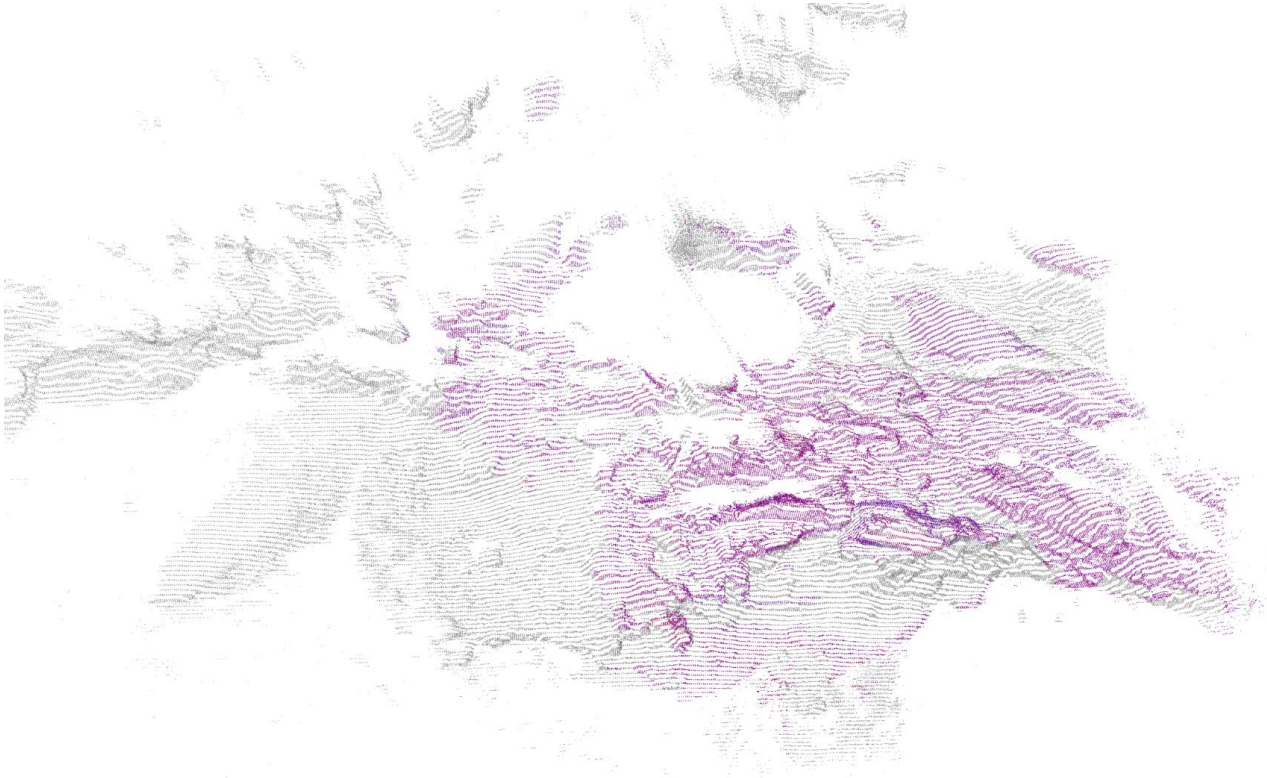
GPS Based Alignment - Top View



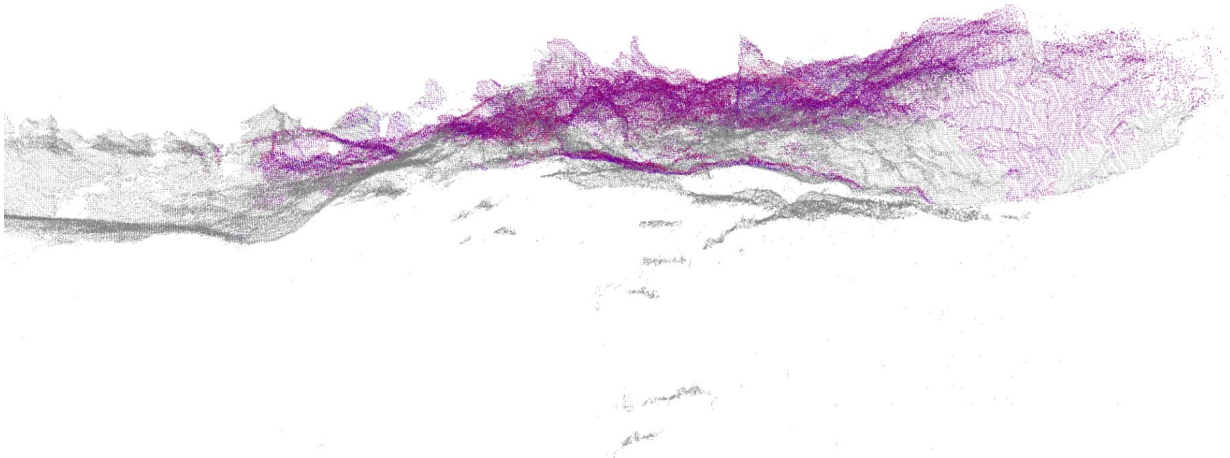
GPS Based Alignment - Side View



Feature Based Alignment - Top View

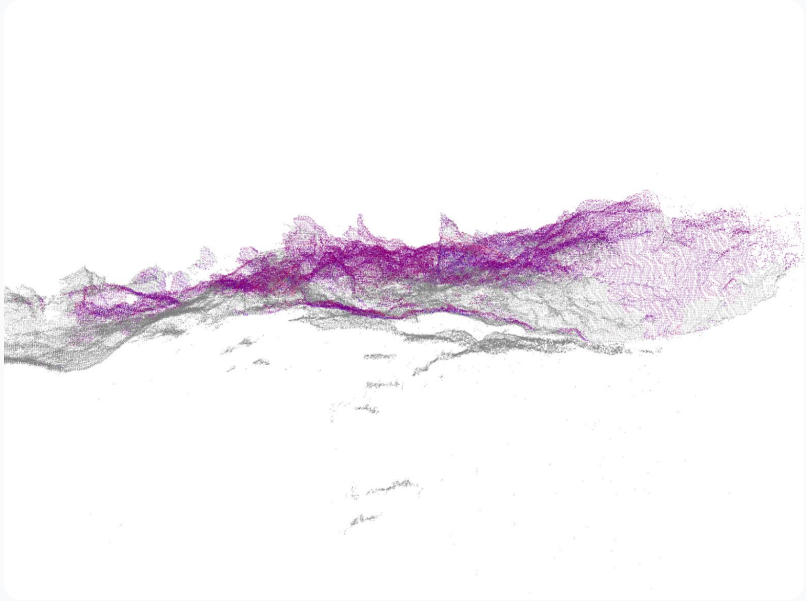


Feature Based Alignment - Side View



Feature-Based Alignment: Comparative Visual Analysis

Side View Comparison



DEM vs M3C2 Volume Calculation

Two different ways to estimate surface change volume from aligned point clouds

M3C2 Volume

- Works directly on the **3D point clouds**.
- For each point, estimate a **local surface normal**.
- Measure signed change **along the normal direction**, not just vertically.
- Captures **local surface-to-surface displacement**, so it handles rough or sloped terrain better.
- Volume is obtained by aggregating these local thickness changes over the area.
- Best when change is not purely vertical and geometry is complex.

DEM Common-Grid Volume

- First converts each epoch into a **DEM / height grid**.
- Uses a shared **XY grid** over the overlap region.
- For each grid cell, compares elevation as **dz = z_2 − z_1**.
- Volume is computed as **Σ(dz × cell area)**.
- Assumes change is mainly represented as **vertical height difference**.
- Simple and intuitive, but more sensitive to gridding, gaps, and vertical bias.

Detailed visualization of feature-based alignment results, highlighting structural fidelity and spatial distribution of matched keypoints.

GPS vs Feature Based Alignment Results

Two volume outputs are computed for each alignment

M3C2 Grid Volume

- Downsample both aligned clouds to 0.5 m.
- Estimate upward target normals.
- For each target point, find nearest source point.
- Signed distance: (epoch2 - epoch1) · normal.
- Accept only correspondences within 1.0 m.
- Bin valid distances into XY grid cells and integrate median per cell.

Results:
GPS M3C2 grid volume: -69.22 m^3
Feature M3C2 grid volume: -63.58 m^3.

DEM Common-Grid Volume

- Build a fixed XY grid over the common footprint.
- For each cell, compute median Z for epoch 1 and epoch 2.
- Compute *dz = median_z_epoch2 - median_z_epoch1*
- Reject cells with too few points or implausible |dz| > 5 m.
- Volume = Σ dz × cell_area.

Results:
GPS DEM volume: -1,393.77 m^3
Feature DEM volume: -1,761.50 m^3.

The DEM volume is sensitive to residual vertical bias and footprint/cell rejection. **M3C2 is more local**; DEM integration is better when the vertical alignment can be trusted.

GPS vs Feature Based Alignment Results

```
Metric                              GPS+TEASER++        Feature
------------------------------------------------------------
M3C2 mean (m)                             0.0202         -0.0125
M3C2 std (m)                              0.3784          0.2851
M3C2 grid volume (m³)                    -69.22          -63.58
DEM volume (m³)                        -1393.77        -1761.50
DEM volume diff (m³)                     367.73
DEM volume diff (%)                       26.4%

============================================================
SAVING
============================================================

  Saved to: /home/hamza-naeem/Documents/project/outputs/alignment_comparison_v2
  Total time: 203.8s
(afr) hamza-naeem@16iax10h:~/Documents/project/Glacial-Loss-Imagery-via-Multi-Pl
```

Key Comparisons

- **Stability:** Feature-based alignment provides lower stable residuals in the current run.
- **Discrepancy:** GPS and feature DEM volumes agree on loss direction but differ by **367.7 m³**.
- **Sensitivity:** The difference indicates residual vertical/footprint sensitivity and should be handled as uncertainty.
- **Validation:** Comparison is an independent check, not a replacement for joint reconstruction when joint SfM is available.

- Used z_offset = median(stable_epoch1_z - stable_epoch2_z) = -4.136m (UTM)

Joint Reconstruction

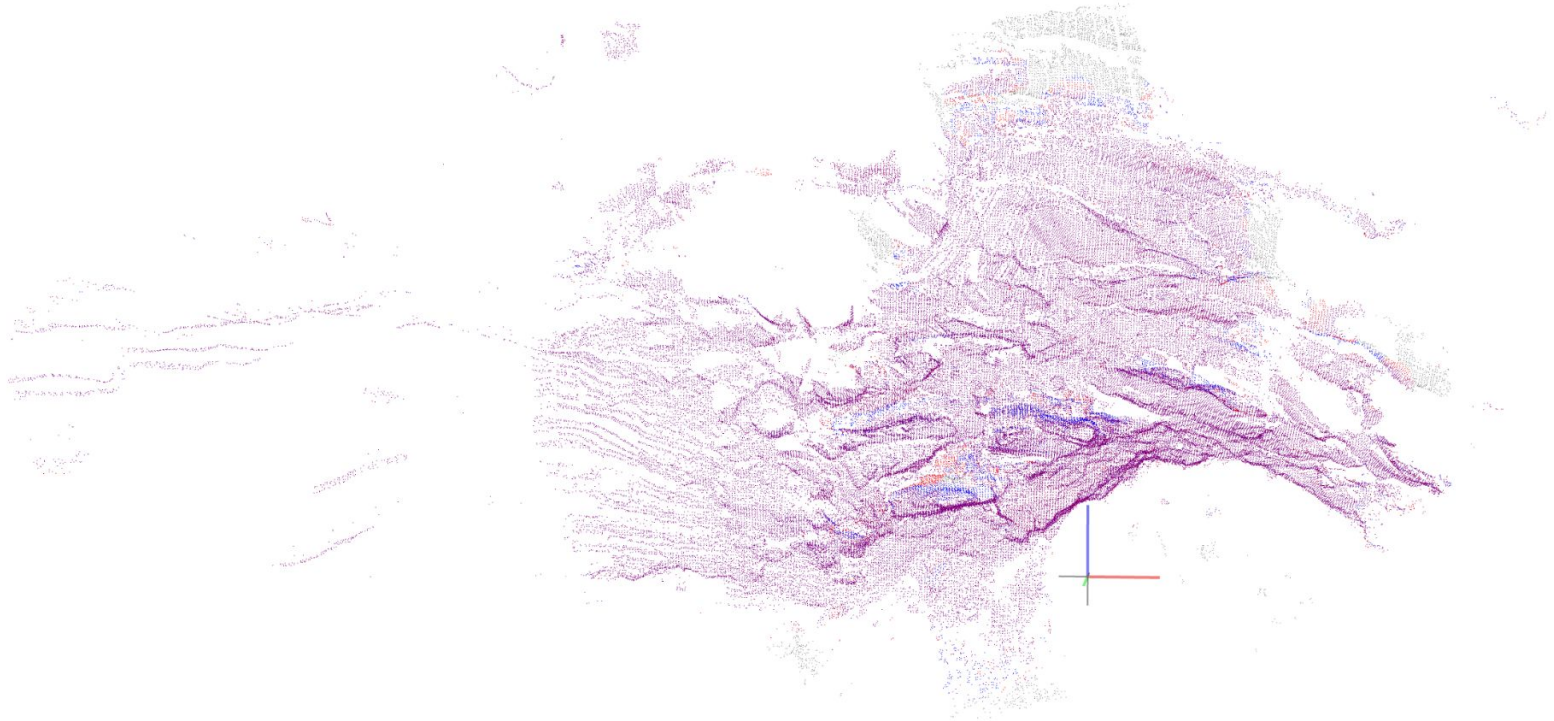
Joint reconstruction pipeline

Both epochs are reconstructed together in one COLMAP model, then split by day. Key benefit: both point clouds already share one local coordinate frame.

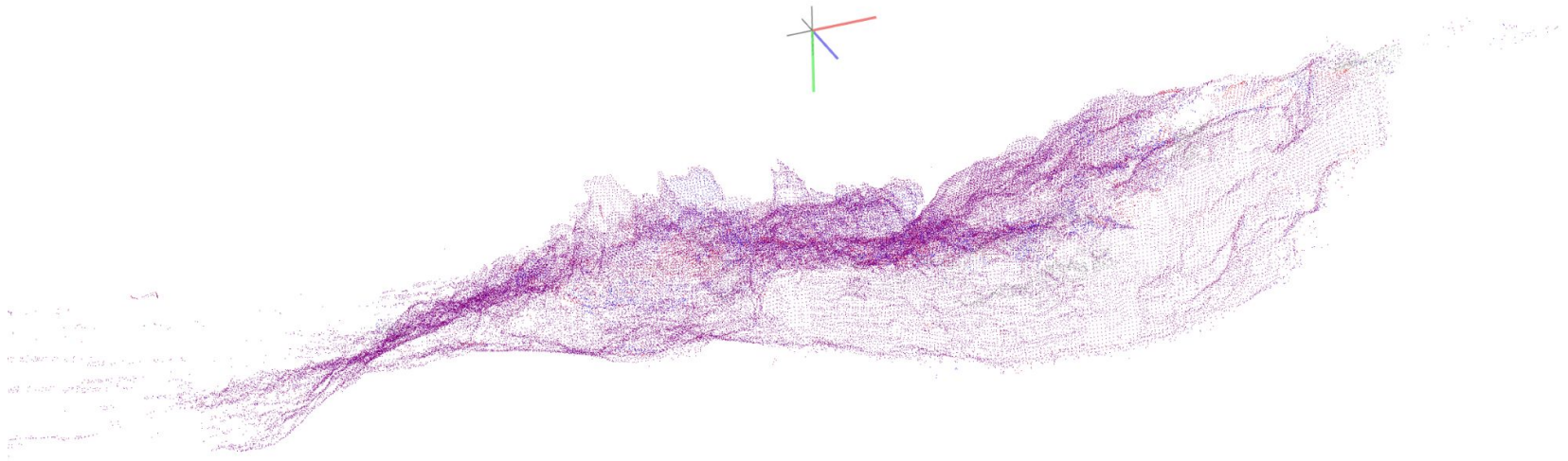
Joint COLMAP reconstruction → Split dense + sparse clouds by day → Sparse validation → Stable region detection → Stable-only GICP → M3C2 + volume

Day 1 Dense Points	Day 2 Dense Points	Metric Scale
1.68M	3.49M	13.124
from Day 1	from Day 2	meters / COLMAP unit from GPS

Joint Reconstruction - Aligned - Top View



Joint Reconstruction - Aligned - Side View



Joint Reconstruction - Day by Day Comparison

August 25th, 2024



August 26th, 2024



Join Reconstruction Process

Stable Region Detection

- Voxel downsample: `0.05 COLMAP`, approximately `0.656 m`.
- Estimate normals for each downsampled cloud.
- Forward nearest neighbor: day1 → day2.
- Backward nearest neighbor: day2 → day1.
- Keep points below the `40th percentile` distance and with normal dot product above `0.7`.
- DBSCAN removes isolated/noisy stable candidates.

Measured Counts

Stage	Day 1	Day 2
Dense input	1,679,227	3,488,660
Downsampled	90,327	104,736
Stable after DBSCAN	33,481	30,463
M3C2 valid target points	100,934 / 104,736	

Join Reconstruction Process

Joint Reconstruction Analysis

Primary Pipeline: Best suited as the primary change-detection pipeline because both epochs share one reconstruction frame.

GPS Robustness: GPS-derived scale is robust: **100/105 GPS inliers, 13.124 m/COLMAP.**

Volume Change: Current M3C2 volume change is small: **-9.21 m^3.**

Next Steps

- Verify these results on a different subset of data
- Can we do change detection in a single flight?
- Joint Reconstruction is better
  - No Issue of Scale
  - But Small Changes are ignored as Outliers in reconstruction
- Need Better Alignment for Separate Reconstruction.
  - Stable regions are very small
  - What if there are no stable regions
- Can we take simple assumptions like same offset for Z ?

Questions

Thank You

Code + data + COLMAP DBs released:
github.com/naeem-ha/Harsh-Feature-Bench
github.com/naeem-ha/Glacial-Loss-Imagery-via-Multi-Platform-Swarm-Evaluation